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Report No: ICR00046

IMPLEMENTATION COMPLETION AND RESULTS REPORT

IDA-62090

ON A

CREDIT

IN THE AMOUNT OF SDR75.5 MILLION

(US$104.89 MILLION EQUIVALENT)

TO THE

People's Republic of Bangladesh

FOR

Sustainable Enterprise Project

August 27, 2024

Environment, Natural Resources & the Blue Economy

South Asia

CURRENCY EQUIVALENTS

(Exchange Rate Effective February 28, 2024)

Currency Unit	=	Bangladeshi Taka (BDT)
BDT109.52	=	US$1
US$0.7539	=	SDR1

FISCAL YEAR
July 1 - June 30

Regional Vice President:	Martin Raiser
Country Director:	Abdoulaye Seck
Regional Director:	Dina Umali-Deininger
Practice Manager:	Christian Albert Peter
Task Team Leader (s):	Eun Joo Allison Yi
ICR Main Contributor:	Sanne Agnete Tikjøb, Ruozi Song, Nina Tsydenova

ABBREVIATIONS AND ACRONYMS

AF	Additional Financing
CCCP	Community Climate Change Project
CCDR	Country Climate Development Report
CPF	Country Partnership Framework
ESF	Environmental and Social Framework
ESP	Environmentally Sustainable Practices
GIS	Geographic Information System
GoB	Government of Bangladesh
GRC	Grievance Redressal Committees
GRM	Grievance Redress Mechanism
HLO	High-Level Outcome
IA	Implementing Agency
ICR	Implementation Completion and Results Report
IDA	International Development Assistance
IRI	Intermediate Indicator
M&E	Monitoring and Evaluation
ME	Microenterprises
MF	Microfinance
MSME	Micro, Small, and Medium-size Enterprises
MTR	Mid-Term Review
OHS	Occupational Health and Safety
OI	Outcome Indicator
PDO	Project Development Objective
PKSF	Palli Karma-Sahayak Foundation
PMU	Project Management Unit
PO	Partner Organization
PPE	Personal Protective Equipment
PPP	Public Private Partnership
RECP	Resource Efficient and Cleaner Production
RF	Results Framework
SEP	Sustainable Enterprise Project
SMART	Sustainable Microenterprise and Resilient Transformation
TOC	Theory of Change
TTL	Task Team Leader
UNFCCC	United Nations Framework Convention on Climate Change
US$	United States Dollars

WB	World Bank
WBG	World Bank Group

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DATA SHEET

BASIC DATA

Product Information

Operation ID	Operation Name
P163250	Sustainable Enterprise Project
Product	**Operation Short Name**
Investment Project Financing (IPF)	Sustainable Enterprise Project
Operation Status	**Approval Fiscal Year**
Closed	2018
Original EA Category	**Current EA Category**
Partial Assessment (B) (Approval package - 29 Mar 2018)	Partial Assessment (B) (Restructuring Data Sheet - 12 Apr 2023)

CLIENTS

Borrower/Recipient	Implementing Agency
The People's Republic of Bangladesh	Palli Karma-Sahayak Foundation (PKSF)

DEVELOPMENT OBJECTIVE

Original Development Objective (Approved as part of Approval Package on 28-Mar-2018)

To increase adoption of environmentally sustainable practices by Targeted Microenterprises.

FINANCING

Financing Source	Original Amount (US$)	Revised Amount (US$)	Actual Disbursed (US$)
World Bank Financing	110,000,000.00	110,000,000.00	104,890,641.53
IDA-62090	110,000,000.00	110,000,000.00	104,890,641.53
Non-World Bank Financing	20,000,000.00	20,000,000.00	20,000,000.00

Borrowing Agency	20,000,000.00	20,000,000.00	20,000,000.00
Total	**130,000,000.00**	**130,000,000.00**	**124,890,641.53**

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Type	Amount Disbursed (US$M)	Key Revisions
12-Apr-2023	Portal	98.14	• Loan Closing Date Extension

KEY DATES

Key Events	Planned Date	Actual Date
Concept Review	30-May-2017	30-May-2017
Decision Review	29-Dec-2017	21-Dec-2017
Authorize Negotiations	25-Feb-2018	14-Feb-2018
Approval	29-Mar-2018	29-Mar-2018
Signing		16-May-2018
Effectiveness	15-May-2018	08-Aug-2018
ICR/NCO		--
Restructuring Sequence.01	Not Applicable	12-Apr-2023
Mid-Term Review No. 01	30-Jun-2021	28-Jun-2021
Operation Closing/Cancellation	29-Feb-2024	29-Feb-2024

RATINGS SUMMARY

Outcome	Bank Performance	M&E Quality
Satisfactory	Satisfactory	Substantial

ISR RATINGS

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US$M)
01	21-Jun-2018	Satisfactory	Satisfactory	0.00
02	10-Dec-2018	Satisfactory	Satisfactory	6.26
03	17-Jun-2019	Satisfactory	Satisfactory	15.77
04	17-Dec-2019	Satisfactory	Satisfactory	26.05
05	30-Jun-2020	Satisfactory	Satisfactory	45.47
06	10-Feb-2021	Satisfactory	Satisfactory	56.09
07	19-Sept-2021	Satisfactory	Satisfactory	80.36
08	06-Feb-2022	Satisfactory	Satisfactory	83.28
09	25-Oct-2022	Satisfactory	Satisfactory	94.66
10	30-May-2023	Satisfactory	Satisfactory	100.53
11	12-Jul-2023	Satisfactory	Satisfactory	100.53
12	14-Mar-2024	Satisfactory	Satisfactory	104.82

SECTORS AND THEMES

Sectors

Major Sector	Sector	%	Adaptation Co-benefits (%)	Mitigation Co-benefits (%)
Industry, Trade and Services	Agricultural markets, commercialization and agri-business	45	35	14
	Manufacturing	45	35	14
	Public Administration - Industry, Trade and Services	10	34	14

Themes

Major Theme	Theme (Level 2)	Theme (Level 3)	%
Environment and Natural Resource Management	Climate change	Adaptation	35
		Mitigation	14
		Air quality management	78

	Environmental Health and Pollution Management	Water Pollution	78
Finance	Financial Infrastructure and Access	MSME Finance	44
Private Sector Development	Enterprise Development	MSME Development	66
Urban and Rural Development	Disaster Risk Management	Flood and Drought Risk Management	35

ADM STAFF

Role	At Approval	At ICR
Practice Manager	Kseniya Lvovsky	Christian Albert Peter
Regional Director	Julia Bucknall	Dina Umali-Deininger
Global Director	Karin Erika Kemper	Valerie Hickey
Practice Group Vice President	Karin Erika Kemper	Juergen Voegele
Country Director	Qimiao Fan	Abdoulaye Seck
Regional Vice President	Annette Dixon	Martin Raiser
ADM Responsible Team Leader	Nadia Sharmin	Eun Joo Allison Yi
Co-Team Leader(s)	Suiko Yoshijima	
ICR Main Contributor	Sanne Agnete Tikjøb, Ruozi Song, Nina Tsydenova	

I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

1. **At the time of appraisal, Bangladesh, a country with 165 million inhabitants, had experienced strong economic growth since the turn of the century, leading to a significant decrease in poverty**. Structural shifts in the economy towards manufacturing and services had created better job opportunities, resulting in millions moving out of poverty and allowing Bangladesh to reach lower-middle income status in 2015. Based on the international poverty line at US$1.90 per day (2011 PPP), the poverty headcount ratio declined from 44.2 percent in 1991 to 13.8 percent in 2016.

2. **Microenterprises, particularly cottage and small-scale businesses, play a critical role in the economy, providing a significant portion of employment opportunities for the poor**. Microenterprises (MEs) were, and remain, vehicles for diversifying economic activity and can make significant contributions to poverty alleviation through self- and wage- employment opportunities. Surveys showed that households with microenterprises had higher per capita income, and MEs accounted for over half of total employment in all enterprises (56 percent or 14 million in 2013).

3. **However, the growing number of microenterprises raised concerns about their combined environmental impact and vulnerability to climate-related risks. Small industries contributed significantly to air and water pollution due to their sheer numbers and use of environmentally inefficient technologies**. While some microenterprises in agriculture and manufacturing produced environmentally sustainable goods, they faced obstacles in reaching high-value markets, adopting cleaner production methods, and accessing quality financing and business registration services. Climate change posed a significant and growing threat to microenterprises across various sectors, and unsafe practices and products, such as contaminated food, not only harmed workers but also impacted consumers. Yet the ME sector had limited focus on environmental sustainability and climate resilience because negative impacts like pollution, health issues, and reduced productivity were often not factored into costs, thereby reducing incentives for microenterprises to transition towards safer and sustainable practices or invest in cleaner technologies.

4. **The Government of Bangladesh (GoB) was dedicated to promoting a greener, cleaner, and climate-resilient economy, but focus had been on export-oriented industries with few initiatives targeting cleaner production practices and technologies in the microenterprise sector**. The sustainable growth of MEs was hindered by various barriers including capacity, market accessibility, knowledge, and financial constraints, and necessitated public sector intervention to address externalities, information asymmetry, capacity gaps, and poor regulation and enforcement. The Palli Karma-Sahayak Foundation (PKSF), an apex microcredit funding and capacity building organization established the GoB in 1991, had been successful in supporting MEs through an ongoing loan program implemented with 178 partner organizations (POs) throughout the country. With PKSF as the implementing agency, the project sought to infuse environmental considerations into financial intermediation by utilizing access to finance for microenterprises to influence their decisions to invest in safer and cleaner technologies and operations.

5. **The World Bank was well positioned to assist the GoB in implementing its strategic development priorities of green growth and climate resilience.** Those priorities were evident in the 2010–2021 National Sustainable Development Strategy (updated in 2013) with its vision of combined economic, social, and environmental goals and the seventh Five-Year Plan for fiscal years 2016–2020 emphasizing pro-poor economic growth while recognizing that in Bangladesh, poverty, growth, and environmental sustainability are inextricably linked. The World Bank could draw on its global technical expertise in economic development, pro-poor policies, and climate adaptation. It also had decades of experience working with the GoB on poverty eradication and strengthening livelihoods and built on the

local demand for microlevel business opportunities demonstrated under the previous World Bank supported Community Climate Change Project (CCCP) Additional Financing (P155106) implemented in partnership with PKSF.

6. **There was a strong rationale for World Bank engagement**. The significant role of microenterprises in enhancing poverty reduction and household well-being in Bangladesh is widely recognized. Thus, by supporting microenterprises to become environmentally friendly and resilient, the project would have broad economic, social, and environmental benefits, and underpinned the twin goals of the World Bank to eradicate poverty and contribute to shared prosperity.

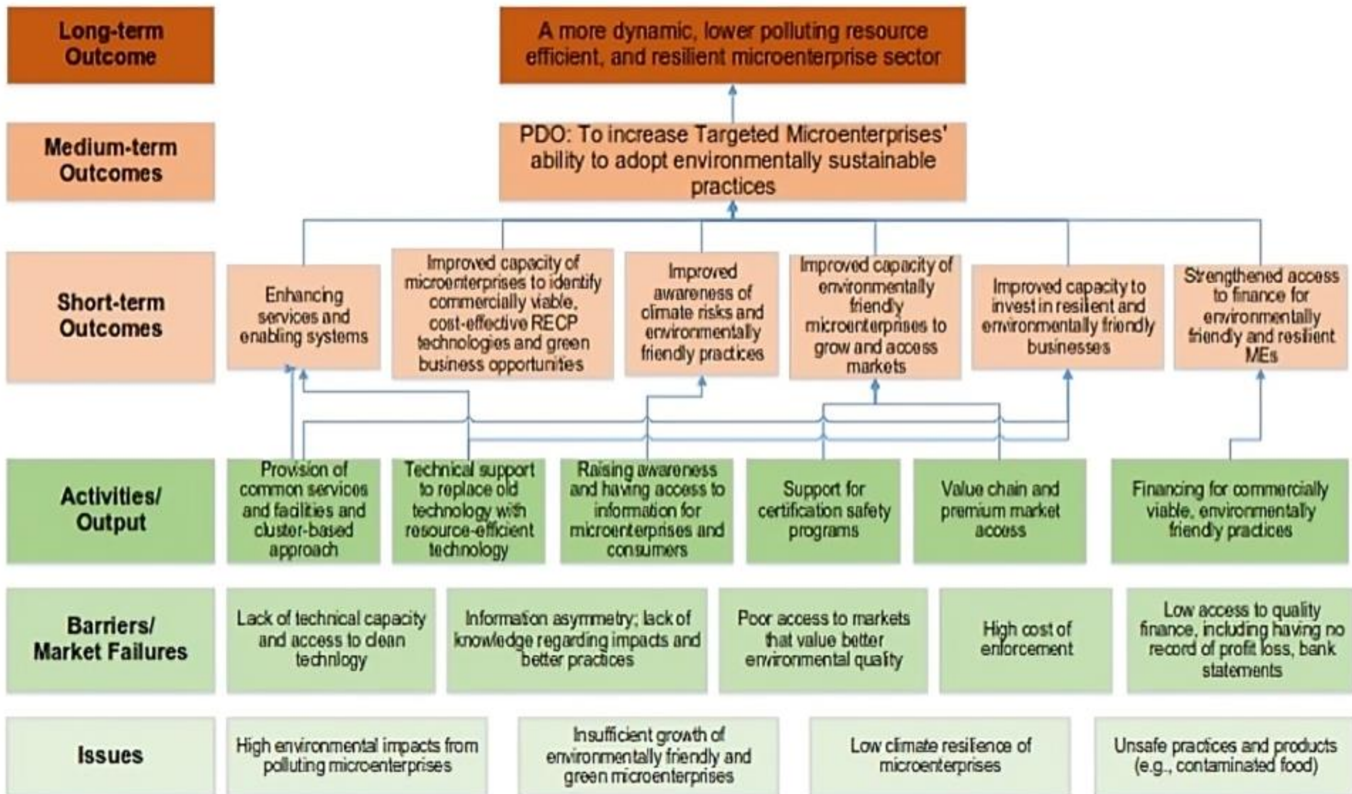
7. **The project was fully aligned with the development priorities outlined in the Country Partnership Framework (CPF) for FY16-FY20 and directly addressed all three pillars of the CPF, namely: growth and competitiveness; social inclusion; and climate and environmental management**. It did so by (i) providing financial and technical support for microenterprises to improve environmental and safety practices and investing in shared service facilities for enhancing profit (CPF objective 1.4: Enhanced business environment and trade facilitation and CPF objective 1.5: Increased financial intermediation); (ii) expanding livelihood opportunities for microenterprises by supporting investment in activities that are environmentally sustainable (CPF objective 2.4: Enhanced rural income opportunities for the poor); and (iii) providing the first level of basic services and health for the intended clients to increase their productivity and ability to take up economic activities in environmentally sensitive, climate vulnerable and disaster-prone areas (CPF objective 3.1: Increased resilience of population to natural disasters in urban and coastal areas and CPF objective 3.2: Improved water resource infrastructure for climate resilience).

Theory of Change (Results Chain)

8. **The project's theory of change built on the need for public sector support to promote investments in environmentally sustainable technologies and practices**. The results chain, developed at the time of appraisal, illustrate how the project sought to address obstacles to adoption of sustainable practices at the ME level in two key sectors, agriculture and manufacturing (Figure 1). Based on a solid analysis of the structural challenges confronting the ME sector to adopt more environmentally friendly practices, the project design offered a comprehensive set of activities, including: raising awareness, providing MEs and consumers with access to information on enhancing Resource Efficient and Cleaner Production (RECP) technologies and processes, offering technical support to replace outdated technology with resource-efficient alternatives, providing financing for commercially viable and environmentally friendly practices, establishing environmentally sustainable common services and facilities in ME clusters, and developing value chains for environmentally sustainable goods with eco-labeling. These activities and outputs were expected to improve the capacity of MEs to invest in commercially viable RECP technologies and green business opportunities, and to develop their value chain and access premium markets for environmentally sustainable goods. Ultimately, these investments were projected to create a more dynamic, lower polluting, resource-efficient, and resilient microenterprise sector and contribute to sustainable economic growth.

9. **Critical assumptions underpinning the TOC**: (i) adequate demand for environmentally sustainable goods produced by the target MEs in premium markets, (ii) the belief that common services and facilities provided under the project would enhance the adoption of environmentally sustainable practices, and (iii) the recognition that MEs owned by women would receive extra attention to facilitate their adoption of environmentally sustainable practices.

FIGURE 1: THEORY OF CHANGE



Project Development Objectives (PDOs)

10. The Project Development Objective (PDO) is to increase the adoption of environmentally sustainable practices by targeted microenterprises (as stated in the Financing Agreement).

- Environmentally sustainable practices were defined as business practices that ensure resource efficiency, low pollution, and/or improved climate resilience.
- Targeted microenterprise included entrepreneurs or microenterprises from the agriculture and manufacturing sectors with legal recognition or registration as well as informal sector operators.

Key Expected Outcomes and Outcome Indicators

11. The project had a single expected outcome, namely to increase the adoption of environmentally sustainable practices by targeted microenterprises. Three related outcomes indicators (OI) were defined:

- Microenterprises targeted by the project that have adopted at least one environmentally sustainable practice (Number)
- Share of target beneficiaries with a rating of “satisfactory” or above on project interventions, disaggregated by sex (Percentage)
- Targeted microenterprises that continue the adopted environmentally sustainable practice, disaggregated by sex of ME owner (Percentage)

Components

12. **Component 1: Enhancing Services and Enabling Systems (Estimated cost: IDA US$24 million. Actual cost: IDA US$21.64 million).** The component sought to enhance resilience, efficiency, and value chain links at the cluster level. Specifically, it aimed to support the clusters to become more environmentally sustainable by enhancing low-polluting, environmentally friendly business practices in clusters, to act as the first step toward introducing environmental sustainability to microenterprises, and to provide better market access to targeted MEs. Subcomponent 1.1 promoted basic safety standards for microenterprises, certified eco-labeled products, built capacity in partner organization, and co-financed grants to invest in common services that have a critical impact on MEs' productivity but are not commercially viable - such as raising a cluster's plinths in a frequently flood-affected area, treating effluents, or providing a waste management and recycling facilities. Subcomponent 1.2 aimed to support capacity building at the ME level. Subcomponent 1.3 sought to support investments in common services that are commercially viable and revenue generating, but essential for making environmentally friendly and green businesses successful.

13. **Component 2: Strengthened Access to Finance for Environmentally Friendly and Resilient Microenterprises (Estimated cost: IDA US$75 million and PKSF US$18 million; total US$93 million. Actual cost: IDA US$75.0 million and PKSF US$18.0 million; total US$93 million**). The component aimed to expand income-generating opportunities for targeted microenterprises by supporting investments in activities that are resource efficient, low polluting, and resilient, while encouraging informal sector participants to obtain formal business registration. PKSF, through its POs, provided financing to microenterprises to adopt environmentally sustainable practices (ESP), while maintaining its standard and well-tested approach for appraising commercial viability and repayment capacity, building on its impressive performance of over 99 percent loan repayment. Funds were split evenly between Subcomponent 2.1, which supported innovative microenterprises to adopt environmentally friendly and resilient green business practices, and Subcomponent 2.2, which supported the expansion of microenterprises that have already adopted those. The project had identified 30 types of priority microenterprise subsectors from the agribusiness and manufacturing sectors that were likely to yield the highest impact.

14. **Component 3: Project Management, Communication and Knowledge Management, and Monitoring and Evaluation (Estimated cost: IDA US$11 million and PKSF US$2 million; total US$13 million. Actual cost: IDA US$6.56 million and PKSF US$2 million; total US$8.56 million).** The component delivered project management and M&E, capacity building of PKSF, and communication and knowledge management. Due to the fluctuation of SDR and US$ rate, the total IDA amount in US$ was reduced which affected the actual costs of Component 3.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION

Revised PDOs and Outcome Targets

15. **The PDO, outcome targets, indicators and components were not revised.**

Other Changes

16. **A level-2 restructuring was approved to extend the loan closing date from June 30, 2023, to February 29, 2024.**

Rationale for Changes and Their Implication on the Original Theory of Change

17. **The project was extended by eight months to compensate for time lost due to COVID-19 induced restrictions (see Section III.B (c)).** The cost-neutral extension of the loan closing date by eight months was required

to complete delayed procurement contracts under component 1 and 3 and realize expected project outcomes. The extension ensured the (i) completion of civil works for the construction of common service facilities, such as mini-Effluent Treatment Plants in MEs clusters, (ii) implementation of training and capacity building of both microenterprises and of PKSF and PO staff, and (iii) finalization of branding and certification activities. At the time of the extension, the project had disbursed US$96 million or 91 percent of the total amount. There were no implications on the original ToC.

II. OUTCOME

A. RELEVANCE OF PDO

Rating: High

Assessment of Relevance of PDOs and Rating

18. **At project closing, the SEP remains highly relevant to Bangladesh's strategic development priorities and is fully aligned with the new CPF for the period FY23-27**. The CPF, anchored in the government's eighth Five Year Plan and the Long-Term Perspective Plan 2021–2041, will support Bangladesh's goal to achieve upper-middle-income country status by 2031 by helping the country to address key barriers to green growth. Central to the growth model embraced by the GoB are three pillars: (i) a competitive private sector to create more and better jobs, (ii) socioeconomic inclusion to expand opportunities for all, and (iii) resilience in the context of climate and environmental vulnerabilities. SEP directly supports these objectives as the ME sector remains critical to the national development agenda. The next phase Sustainable Microenterprise and Resilient Transformation (SMART) project builds directly on the achievements and lessons learned of the SEP, providing additional evidence of the continued relevance of SEP objectives.

19. **In the context of the CPF, SEP contributes to several objectives**. Objective 1 (Improved business environment for broad-based private sector) as it promotes green, energy-efficient, cleaner businesses at the ME level that are socially inclusive and enhance competitiveness through green/eco-labelling of products. SEP also contributes to Objective 5 (Enhanced economic opportunities for women and vulnerable groups) by strengthening and expanding livelihood opportunities for MEs by supporting investment in activities that are environmentally sustainable and specifically targeted to women. SEP specifically contributes to Objective 8 (Enhanced sustainability and productivity in the use of natural capital for climate-smart green growth) through adoption and promotion of climate resilient RECP business practices and greener technologies.

20. **The Country Climate Development Report (CCDR) was endorsed by Bangladesh in November 2023 at UNFCCC COP28 and confirms the continued high relevance of SEP's development objectives**. The CCDR highlights the importance of transitioning to a green, resilient, inclusive growth model focusing on greener infrastructure planning and development as drivers of productivity and growth. The project contributes to Priority Area 1: People, Climate-Smart Spatial Development, which leverages financing mechanisms for locally-led climate action, increases awareness for local-level climate actions and strengthens systems for adaptation. The project also supports Priority Area 2: Delivering development benefits with decarbonization, focusing on energy efficiency and circular economy solutions in garments and textile factories.

21. **The project has also stayed highly relevant to the World Bank's evolving corporate agenda and is fully aligned with new strategic initiatives.** The project is aligned with the new World Bank Evolution Roadmap and its mission of ending extreme poverty and boosting prosperity on a livable planet through enhanced environment for increased resilience. It supports the WBG's recent Green Resilient and Inclusive Development agenda and is consistent with the WBG's Climate Change Action Plan 2021–2025. The project is also consistent with the World Bank's

Maximizing Finance for Development strategy, as the project provides a liquidity injection that will benefit MEs and has the potential to support further business creation in the future.

B. ACHIEVEMENT OF PDOs (EFFICACY)

Rating: Substantial

Assessment of Achievement of Each Objective/Outcome

22. **Project achievements are assessed against the single development outcome, which is 'to increase the adoption of environmentally sustainable practices by targeted microenterprises.'** The data for assessing the outcome is drawn from PKSFs M&E system as well as the Mid-term Evaluation and the Final Evaluation Survey done by a third-party organization. The assessment is further informed by the Borrower's Implementation Completion and Results Report, the latest ISR, and by information gathered during the ICR mission and from the Project Management Unit (PMU).

23. **The efficacy section is organized in three parts:** a) adoption of environmentally sustainable practices by targeted microenterprises, b) impact of adopted environmentally sustainable practices on microenterprises, and c) continuation of adopted environmentally sustainable practices by targeted microenterprises**.**

a) Adoption of environmentally sustainable practices by targeted microenterprises

24. **The project has increased the adoption of environmentally sustainable practices among beneficiary microenterprises as evidenced by a 75 percent adoption rate (Table 1), compared with a target of 50 percent.** A total of 64 sub-projects were approved for implementation by 47 Partner organizations with a combined envelope of US$82 million (including both US$69 million from IDA and US$18 million from PKSF). The funds benefited 65,124 microenterprises, who obtained microloans to invest in ESPs as means to protect the environment, strengthen the resilience of their business, and improve sales through green certification and access to premium markets. At completion, out of the 65,124 MEs that signed loan agreements, 48,845 ME (or 75 percent) have adopted at least one of twenty ESPs eligible for financing. This compares favorably to the original target of a 50 percent adoption rate, which is how OI 1 was defined (OI 1 / IRI 3 in Table 1 below, see also the Results Framework (RF) in Annex 1).

25. **On average, each ME adopted approximately three environmentally sustainable practices out of 20 eligible practices (see the list of eligible and adopted ESPs in Table 11 in Annex 7).** Many of the practices are cross-cutting in nature with co-benefits for improving resource efficiency, reducing pollution, and building climate resilience, such as installation of solar panels, air circulation systems, and improved electrical wiring, as well as use of natural and organic fertilizer and dye. Other ESPs focused on improving occupational health and safety (OHS) first, before introducing more advanced technologies to green production processes, including installation of first aid boxes, fire management tools, and proper sanitation facilities as well as use of personal protective gear. As eligible practices were only defined during implementation and in consultation with the beneficiary POs and MEs, this was considered a natural evolution in MEs general awareness of safer and greener manufacturing processes (see also Section III).

Table 1: Indicators relevant to adoption of environmentally sustainable practices among targeted MEs

#	Indicator	Target	Actual	% Achieved
OI 1	Microenterprises targeted by the project that have adopted at least one environmentally sustainable practice (#)	20,000	48,845	244%
IRI 2	Share of targeted microenterprises provided with capacity building support to adopt environmentally sustainable practices (%)	70	91	130%
IRI 3	Microenterprises that sign loan agreements with POs under the project (#)	40,000	65,124	162%

26. **The adoption rate of environmentally sustainable practices is markedly higher among beneficiary MEs than among other MEs in the targeted clusters (Table 2), but the project likely had a positive spill-over effect on ESP uptake in non-project MEs within the same clusters.** The project had a positive impact on the ESP adoption rate among beneficiary MEs. The Final Evaluation Survey showed that 80 percent of beneficiary MEs and 49 percent of MEs in the control group had adopted ESPs over the same period. The finding indicates that the project caused a 30 percentage point higher uptake of ESPs among beneficiary MEs. It also likely signals the positive spill-over effect that the project had on other MEs within the same clusters as it sought to raise awareness and establish shared facilities to address waste management, effluent treatment, micro-storage needs, etc. to promote resource-sharing through a common infrastructure. Without geotagged micro-level data, it is not possible to prove a spill-over effect from beneficiary MEs to other MEs in close proximity. However, it is safe to assume that data on ESP uptake in the control group cannot be extrapolated to all loan-taking MEs in Bangladesh. It can therefore be inferred that there is a strong relation between MEs in close geographical proximity and that behavioral practices is easily spread when families and businesses interact at the ME level. Under the next-phase SMART project, this is being addressed through improved data collection practices with geotagged location entries (see also Section V).

Table 2: Adoption of environmentally sustainable practice by microenterprises in targeted clusters (Final Evaluation Survey)

Adoption of at least one ESP	Beneficiary Group			Control Group		
	Male (%)	Female (%)	Total (%)	Male (%)	Female (%)	Total (%)
Yes	80.6	78.7	79.7	50.4	45.1	48.5
No	19.4	21.3	20.3	49.6	54.9	51.5
Sample size (#)	1,994	1,635	3,629	1,412	812	2,224

b) Impact of adopted environmentally sustainable practices on microenterprises

27. **Between 60 to 90 percent of all MEs in targeted clusters, who have invested in ESP with SEP-funds or other funds, report moderate to significant improvements across eleven dimensions of well-being.** Across both beneficiary and control groups, respondents reported experiencing moderate to significant improvements from ESP investments on dimensions of livelihood, employment, education, health, production, income, savings, purchasing capacity, assets, investments, consumption, quality of life, psychological well-being, and social well-being. Annex 8 on Project Beneficiaries includes graphs from the Final Evaluation Survey as well as case studies showcasing the impact of ESP investments on microenterprises and local communities.

28. **Among beneficiary MEs, 66 percent are "satisfied" or "highly satisfied" with the project (Table 3**). The achievement, which was calculated as a composite index based on beneficiaries' responses to 11 dimensions of project performance, highlights the positive impact that the project had on microenterprises participating in SEP.

Table 3: Indicators relevant to the impact of adopted environmentally sustainable practices among beneficiary MEs

#	Indicator	Target	Actual	% Achieved
OI 2	Share of target beneficiaries with a rating of "satisfactory" or above on project interventions (%)	60	66	110%

c) Continuation of adopted environmentally sustainable practices by targeted microenterprises

29. **The majority of microenterprises who invested in and adopted environmentally sustainable practices continued the practice for longer than 12 months**. This finding from the Final Evaluation Survey is true for beneficiar

and for MEs in the control group, as both groups report above 90 percent continuation rates of adopted ESPs (Table reported in the Final Evaluation Survey, this testifies to the beneficial impact of environmentally sustainable practices c profitability through more resilient business practices to climate events and through recognized environmental certifi advanced product placement and access to premium markets.

Table 4: Duration of adopted ESPs among beneficiary MEs and MEs in the control group (Final Beneficiary Survey)

Duration of adopted ESPs	Beneficiary Group			Control Group		
	Male (%)	Female (%)	Total (%)	Male (%)	Female (%)	Total (%)
<3 months	1.3	1.2	1.3	3.2	2.5	3.0
3-5 months	1.7	2.3	1.9	5.5	2.5	4.5
6-12 months	1.6	1.9	1.7	1.7	1.4	1.6
12+ months	95.4	94.6	95.0	89.6	93.7	91.0
Sample size (#)	1 613	1 313	2 926	712	366	1 078

30. **Key to sustaining the practices adopted, was the project's focus on building an enabling environment for microenterprises to institutionalize the practices through support for common infrastructure facilities and for MEs certification, branding, and access to premium markets**. First, SEP established 25 revenue-generating and 25 non-revenue-generating common facilities in each cluster to enhance MEs growth, competitiveness and productivity. Facilities established included, among others: i) waste management and recycling facilities to manage waste and effluent water, which pollutes potable water sources and threatens the health of humans and livestock, ii) sustainable farming to strengthen crop resilience to extreme weather events and address food insecurity, and iii) enhanced dairy, horticulture and dry fish processing businesses to develop value-added industries. The common facilities are available to all MEs in the clusters, acting as an ongoing multiplier for the benefits of RECP and ESP practices. See Table 17 and Table 18 in Annex 7 for a full list of common facilities established.

31. **Second, the project helped increase the capacity of MEs on eco-labeling and access to premium market by providing support for business and product certification and developing branding and communication material**. Through a comprehensive effort to build the capacity of POs and MEs (see Table 12, Table 13, and Table 14 in Annex 7), coupled with extensive communication, branding and marketing campaigns (see also outputs by components in Annex 1.B), SEP facilitated the formal registration of 12,000 microenterprises (nearly 20 percent of MEs), the certification of products with relevant authorities for 1,246 MEs, and created brand identity and related assets, such as logos, packaging, and production support, for MEs working with 37 POs in 15 different sectors. Combined, these efforts have helped develop the foundation for sustaining and expanding environmentally sustainable practices among microenterprises in targeted clusters (see Table 5).

Table 5: Indicators relevant to the continuation of adopted environmentally sustainable practices among targeted MEs

#	Indicator	Target	Actual	% Achieved
OI 3	Targeted MEs that continue the adopted environmentally sustainable practice (%)	40	95	238%
IRI 1	Micro-enterprise clusters provided with common services (#)	25	25	100%

Justification of Overall Efficacy Rating

32. **Overall project efficacy is rated Substantial. The project achieved its development objective and exceeded the targets of all three PDO indicators**. It successfully increased the adoption of environmentally sustainable practices in targeted microenterprises and built an enabling environment through common facilities, brand development and

market expansion, which can help create positive spill-over effects to other MEs and ME clusters. However, not all ESPs adopted were geared towards RECP outcomes as conditions on the ground demanded that considerations for OHS first be addressed. The outcomes have transformed communities (see case studies in Annex 8) and improved recognition of MEs products and businesses, which is the best foundation for continued practice and expanded adoption of RECP and ESP among microenterprises. The results also constitute the baseline for scaling-up successful activities through the follow-up Sustainable Microenterprise and Resilient Transformation (SMART) (P178996) project.

C. EFFICIENCY

Rating: Substantial

Assessment of Efficiency and Rating

33. **The innovative character of the project design coupled with solid implementation arrangements and nimble supervision were key factors in ensuring a substantial level of efficiency**. As the first project of its kind in the World Bank, the ambitious nature of the project envisioned using a financial mechanism to improve environmental sustainability among MEs. The IA's previous significant experience with World Bank projects provided a solid foundation for managing the implementation of project activities. During implementation, pointed project management enabled several adaptations, which increased efficiency, including using OHS practices as an entry point to improving environmental sustainability practices (see Section III) and adopting "nudge theory" to incentivize a positive shift in ME behavior (see Section V). At completion, the documented impact of ESP investments on a range of social and economic parameter are evident in the final beneficiary survey. Despite the global pandemic and some delays with procurement, there were no cost overruns.

34. **A comparison of actual cost by component to appraisal estimates indicate a reasonable level of efficiency**. Component 1 and 2 were financed at 90 percent and 100 percent, respectively, while actual expenditure under Component 3 was a little more than half the appraisal estimate (60 percent). Due to fluctuation of the exchange rate between SDR and USrate, the total actual IDA contribution was app. US$104.82 million. The difference was covered by PKSF's contribution, which was 100 percent disbursed at BDT1,607 (equal to US$20 million) over the project lifetime.

35. **Overall project efficiency is rated Substantial**. It included a robust economic and financial analysis at appraisal. The update of the economic and financial analysis in this ICR preserves the methodology and basic parameters adopted during the project preparation, but to the extent possible, uses actual data gathered as part of M&E and adjusts forecasts accordingly. The methodology is summarized in Annex 4.

36. **The analysis indicates that the project has an IRR of 7 percent with an NPV of US$27.47 million without considering environmental benefits**. The NPV ranges from US$52.39 million to US$60.81 million depending on the shadow price of carbon when environmental benefits are included. Results are summarized in Table 6. The economic benefits are driven by the increase in MEs' profits due to SEP.

Table 6: Estimated costs and benefits of the project at completion

	NPV (million dollars)	B/C Ratio	IRR
Without SPC	29.04	1.24	7%
With the low value of SPC	37.06	1.31	9%
With the high value of SPC	45.30	1.38	11%

D. JUSTIFICATION OF OVERALL OUTCOME RATING

37. **The overall outcome rating is Satisfactory**. The project's relevance rating is high, efficacy rating is substantial, and efficiency rating is substantial. The project remains relevant to the GoB's development agenda and

is well-aligned with Covid-19 related response and recovery measures, especially with regards to promotion of MEs. The project has achieved and surpassed the targets of almost all the indicators. The project's costs relative to outcomes achieved far exceed estimates at appraisal. SEP stands out as a pioneering initiative that merges business growth with the integration of environmentally sustainable practices. The low cost of implementation, coupled with substantial environmental and economic gains, encouraged a significant uptake of ESPs among microenterprises and exceeded the initial targets in the RF.

E. OTHER OUTCOMES AND IMPACTS

Gender

38. **The project was gender tagged and the design sought to address existing gender gaps by improving women's empowerment through access to finance for green investments with specific indicators included in the RF to track outcomes.** Women were the primary beneficiaries of the project with 84 percent of loan-takers being women-owned MEs (IRI 3). In Bangladesh, only 35 percent of women are part of the workforce (IFC, 2016). At PKSF, while the large majority of ME loan-takers are women, only 22 percent of the women own their business (PAD). This puts project achievements in perspective. By cultivating a focus on women's participation, the target for women-owned ME loan-takers was overachieved by 280 percent. In the Final Evaluation Survey, women responded to questions on project performance, provision of training and capacity building, and sustainability of adopted practices with the same level of satisfaction as male respondents, thereby overachieving on all indicators tracking results for female beneficiaries.

Table 7: Gender disaggregated indicators

#	Indicator	Target	Actual	% Achieved
OI 2	Share of target beneficiaries with a rating of "satisfactory" or above on project interventions – percentage female (%)	30	65	217%
OI 3	Targeted micro-enterprises that continue the adopted environmentally sustainable practice – percentage female-owned (%)	30	95	316%
IRI 2	Share of targeted microenterprises provided with capacity building support to adopt environmentally sustainable practices – percentage female-owned (%)	30	52	173%
IRI 3	Microenterprises that sign loan agreements with POs under the project – percentage female-owned (%)	30	84	280%

Institutional Strengthening

39. **Over the past 13 years, PKSF and its Partner Organizations have undergone significant institutional strengthening through their roles as implementing agencies for various projects, beginning with the World Bank-financed CCCP valued at US$12.5 million.** This initial project laid the groundwork for subsequent initiatives such as the Sustainable Enterprise Project (US$110 million) and the Sustainable Microenterprise and Resilient Transformation Project (US$250 million). Through extensive capacity building and hands-on experience, particularly in addressing environmental and social issues for microenterprises, PKSF and its partners have experienced a steep learning curve, enhancing their ability to manage and execute complex development projects effectively.

40. **Developing a strong operating framework for environmental management within PKSF was central to the attainment of project objectives to sustainably facilitate adoption of ESPs at MEs level.** The opportunity to support MEs and POs in adopting ESPs strengthened the Environment and Climate Change Unit (ECCU) within PKSF and enabled the establishment of an Environmental Management System in PKSF. The ECCU has played a pivotal role in advancing environmental and social governance within SEP sub-projects. By offering technical and advisory support, the ECCU has guided the SEP-PMU in the adoption and execution of the Environmental and Social Management Framework (ESMF). Furthermore, the development and distribution of 16 tailored environmental and social

management guidelines have been instrumental for POs to align with best practices in implementing ESMF effectively. The hands-on approach, including field visits, underscores the ECCU's dedication to providing comprehensive technical support to POs, fostering a sustainable and responsible project management environment.

Mobilizing Private Sector Financing

N/A

Poverty Reduction and Shared Prosperity

41. **The project has been a catalyst for economic growth, leading to notable improvements in investment and employment, which are essential for boosting household incomes and alleviating poverty.** An analysis within the SEP framework indicates beneficial effects on the economic status of targeted showing a positive trajectory in income levels. Furthermore, the value of both non-durable and durable assets of targeted MEs has risen. Additionally, there has been a considerable rise in sales revenue across various sectors, where MEs showed a greater increase in business investment over the past three years. Targeted MEs were also more inclined to reinvest profits to expand business operations, indicating a robust cycle of investment and growth.

Other Unintended Outcomes and Impacts

42. **SEP played an unexpected role in the effort to respond to the COVID-19 crisis.** Under guidance by the project, mini-garment manufacturers quickly repurposed their production to start producing masks and PPEs. In total, 2.2 million masks and 9,000 PPEs were manufactured by SEP supported MEs. PKSF encouraged other micro-enterprises to step up as well and as a result, 82.2 million masks and 33,000 PPEs were manufactured through various PKSF-managed programs. As such, MEs not only managed to pay salaries for the workers when other businesses could not, but also served the country during the pandemic.

III. KEY FACTORS AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

43. **Project objectives responded directly to GoB strategic development objectives and incorporated key lessons learned from the CCCP and other projects**. Actual project preparation took just ten months but was rooted in the discussions and lessons stemming from the previous CCCP implemented in partnership with PKSF. Through on-going dialogue, the key development challenges related to environmental impacts of microenterprises were clarified and solutions were proposed to offer financing for green investments to MEs to promote growth, jobs, and sustainable development. The project team drew on the demonstration effects delivered by CCCP, such as local demand for micro-level business opportunities, new opportunities for women to enter the workforce, and the capacity of local communities to become more financially self-reliant and resilient to extreme weather and climate risks. In addition, the Kenya Micro, Small, and Medium Enterprise (MSME) Competitiveness Project (P085007) showed that MEs need an integrated package of interventions (access to finance, improvements in the business environment, and business development services that build capacity) to address interrelated constraints and enable microenterprises to access markets on a sustainable basis. Finally, the Additional Financing for Financing Energy Efficiency at MSMEs Project (P158033) in India demonstrated the need for a financial scheme to promote environmentally friendly investments in MSMEs. Building on these insights, the SEP project incorporated these lessons into the project design.

44. **SEP, as the first environment sector project to target MEs, was ambitious in introducing a new approach to addressing environmental challenges while keeping the objectives realistic.** Without other benchmark projects

in the World Bank to compare with, SEP was designed to test if environmental sustainability could be introduced and sustained at the microenterprise level. Without regulation to pressure MEs to perform better environmentally and factor in the cost of environmental externalities, MEs had, unlike bigger companies and industries with global value chains, little incentive to invest in safer and greener technologies and infrastructure. By focusing on proving economic viability and environmental sustainability of microfinance for green investments first, the plan was to broaden the scope in a second phase to address compliance, enforcement, and policy development. This vision for the second phase is now being realized in the follow-on SMART project.

45. **The project benefited from strong internal logic between expected outcomes, components, and planned activities.** A basic ToC was prepared, which included a diagnosis of the structural barriers and market failures that prevented MEs adoption of better and cleaner practices. A sound understanding of the structural challenges helped target the design of realistic responses. With just two main components, one focused on improving clusters and the other focused on on-lending to MEs, the design was kept simple. The results framework was appropriately geared towards tracking adoption of environmentally friendly technologies in MEs rather than assessing the environmental impacts of those investments. This allowed the project to stay focused on demonstrating the value and viability of green microfinance before considering the environmental outcomes of realized sub-project. This served the project well in terms of achieving intended outcomes, but still leaves the larger question unanswered on how these sub-projects perform environmentally.

46. **Key to the successful achievement of project outcomes was a well-established institutional infrastructure.** The implementing agency was a reliable and experienced partner, who had worked with the World Bank for decades and had a strong performance record. On one hand, this was the first microfinance project in the environment sector team for the World Bank, while on the other hand this was the first environmental management project for the implementing agency. This required a high level of trust and close collaboration between the World Bank team and the implementing agency. A financial performance review indicated that PKSF was a viable financial institution and qualified to act as a wholesale organization for on-lending IDA funds to the microfinance sector as required under OP 10. Furthermore, the IA had a comprehensive M&E system in place to monitor sub-projects with a proven record of keeping implementation on track.

47. **At approval, project preparation was at an advanced stage of readiness for implementation with overall project risk rated Substantial and key mitigation measures based on engagement and monitoring.** Identified risks, including technical, institutional, and fiduciary, were all rated Substantial. Arrangements for mitigating these risks relied on (i) built-in multi-layer monitoring to introduce results-based monitoring and GIS monitoring under Component 3, (ii) highly rated POs according to PKSF's own assessment and classification of its partner organizations, and (iii) engagement of targeted MEs and related business clusters through consultative processes and feedback mechanisms. Attention was given to managing reputational risks with clear communication and transparency, given the reliance on NGOs to implement project activities and general exposure to the public through microenterprises. Third-party monitoring of project activities was planned at project start, mid-term and completion to ensure transparency and enable feedback loops.

B. KEY FACTORS DURING IMPLEMENTATION

a) Factors subject to the control of the government and/or implementing entities

48. **The project enjoyed strong and continuous commitment from the GoB. The GoB showed continuous support to PKSF and to the project and remains committed to developing the ME sector as a means to achieve green growth and climate resilience**. The GoB contributed US$18 million in counterpart funding for loans to MEs (Comp. 2) and provided US$2 million to cover operating costs for the implementation of the project (Comp. 3), such as payment of government employees and recurring expenses (office and technical equipment, remuneration,

vehicle maintenance and repair, etc.). The combined amount represents a mobilization rate of 100 percent against the expected contribution at appraisal.

49. **Solid institutional commitment and capacity to implement the project is one of the key factors leading to the successful achievement of intended development objectives.** Project implementation benefited from PKSF's twenty years of experience working on microenterprise development and solid familiarity with World Bank policies and standards. The PMU was adequately staffed with capable specialists, who implemented the project in compliance with environmental and social safeguards requirements, and financial and fiduciary management throughout the project period. The project was the first of its kind implemented by PKSF that addressed the environmental impact of microenterprises without jeopardizing its principle of profitability. The partner organizations were also instrumental in terms of mobilizing, motivating, capacity building, and nudging the microenterprises to adopt environmentally sustainable practices.

50. **During the early implementation phase and the onset of the global pandemic, when eligible ESPs were being defined in consultation with stakeholders, it became clear that the path towards environmental sustainability in the ME sector required a practical approach where OHS measures were addressed as a way to create awareness and a solid foundation for promoting greener production processes.** The premise was that by first addressing occupational health and safety measures, the project would help identify the opportunities for evolving MEs business practices towards adoption of more advanced technologies for resource-efficient use of raw and waste material and for cleaner production processes and pollution control (see eligible ESPs in Table 11 in Annex 7). Rather than a change of approach or a revision of eligible ESPs, this was a practical adaptation to meet ME's needs on the ground. Furthermore, many of the OHS practices held important co-benefits for building climate resilience.

51. **To motivate the MEs and increase the adoption rate of environmental sustainability and safety practices, the PMU developed a communication strategy and two-year work plan based on "nudge theory".** The project strategically tested various nudging techniques aimed to create a positive shift in MEs business behavior. Through a combination of positive reinforcement, recognition, and information dissemination, nudging strategies were designed to inspire a culture of responsibility and sustainability within the microenterprise sector. Strategies included identifying model microenterprises, recognizing best-in-class MEs in different categories, introducing an environmental and social credit rating system, developing a Green Credit Rating System to earn badges for implementing eco-friendly practices, peer learning, and case studies of successful MEs. The combination of these nudging strategies proved effective in instilling a sense of responsibility and sustainability within the micro-enterprise community in Bangladesh, as evident in the overachievement of the adoption and continuation rate of environmental practices. Going forward, a better understanding of the specific context and needs of the target audience will continue to guide the evolution of nudging interventions.

b) Factors subject to the control of the World Bank

52. **Overall, the World Bank team provided timely supervision and solutions to meet implementation challenges**. The Borrower ICR specifically highlights the regularity of coordination meetings with the World Bank team and the dedicated effort to anticipate implementation challenges at the sub-project level to advance implementation progress and ensure the attainment of development outcomes. The main decisions that positively affected project implementation and outcomes include extending the closing date eight months to address an implementation delay caused by the global pandemic and advancing discussions that led to a phase two project. The project was rated satisfactory throughout the implementation period, from effectiveness to closing.

53. **The mid-term review was a positive milestone, which planted the seeds for a follow-on project**. At the time of the MTR in May 2021, the project implementation period was 61 percent complete, a total of 35,107 MEs

(88 percent of the target) had signed loan agreements, of which 83 percent were signed by women, and disbursement had reached US$76.8 million, exceeding 70 percent of the approved credit. Based on the impressive early achievements, the PKSF through the Ministry of Finance (MoF) submitted a request for additional financing of US$250 million in May 2020 to cover a new sub-sector and further promote the generation of new services to help MEs address economic and environmental barriers. The World Bank team explored the possibility of additional financing (AF) as a way to build on SEP's early success and to build back better and greener post-COVID. However, due to the introduction of the World Bank Environmental and Social Framework (ESF) , projects prepared under old safeguards policies could not process an AF through restructuring (only AFs due to cost over runs were processed) and required a new project to be created.

54. **The process of approving the sub-projects was lengthy and required two rounds of reviews by the PKSF and the World Bank each**. This rigorous process was followed to award each sub-project and ensured a high level of quality control. It is one of the key factors leading to the overachievement of the adoption rate of environmental practices. However, the process also caused delays as the reviews sometimes took longer than expected, though not enough to affect the overall project outcome.

c) Factors beyond the control of the government and/or implementing entities

55. **The Covid-19 pandemic caused implementation and procurement delays, but the Bank team acted swiftly to extend the project closing date to complete severely affected project-related activities.** Field visits by the PMU were initially interrupted due to COVID-19, but as the situation improved, regular visits resumed to monitor financial management activities and validate transactions at the PO level. Project implementation faced delays due to COVID-19, prompting the client to request an 8 month no-cost extension to fulfill activities necessary for achieving the PDO. The pandemic also caused procurement delays, impacting both the initiation and implementation of contracts. The MTR highlighted slow procurement progress for planned studies due to COVID-induced restrictions, leading the World Bank team to request the PMU to review and update procurement items and provide a timeline for timely completion. In response to pandemic challenges, the Bank extended the project closing date to accommodate severely affected activities.

IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

Rating: Substantial

M&E Design

56**.** **The TOC was coherent with a clear expression of causality between objectives, activities, outputs and outcomes.** The design of component activities supported the PDO in an integrated approach, with part of Component 3 dedicated to independent monitoring and verification of project activities and outputs by an independent third-party. By design, the RF had an emphasis on collecting gender-disaggregated data.

57**.** **The results framework was sufficient to guide SEP in tracking its achievements towards attaining project development objectives.** At appraisal, the PDO and RF were deliberately designed to measure results at the output level given the contextual challenges to assess environmental impacts due to the large diffusion and small scale of project supported activities. Because it was known already at the appraisal stage that the project would not be able to demonstrate – through meaningful monitoring and evaluation – the expected positive environmental impacts of adopted practices, both the PDO and RF were structured at the output level with a measure of sustainability. This ensured strong cohesion between the PDO and related indicators, and as such the RF was appropriately geared to assess project objectives. A number of case studies were developed to provide anecdotal evidence of positive environmental outcomes (see Annex 8).

58. **M&E arrangements were integrated into the existing PKSF system.** At the time of appraisal, PKSF had extended microfinance loans to 1.3 million borrowers across the country. SEP M&E was designed to benefit from the existing M&E system in place at PKSF with a dedicated M&E officer. PKSF had the overall responsibility for data collection and upstream progress reporting to the Steering Committee and the World Bank on an annual basis. Meanwhile, the POs were each in charge of collecting regular M&E data from MEs on their respective sub-project activities to allow for regular assessment of project implementation progress.

M&E Implementation

59. **M&E was rated 'highly satisfactory' throughout implementation**. The results framework was not revised, though in hindsight given the overachievement of key indicators it could have been appropriate for the team to adjust the targets upwards.

60. **M&E was carefully integrated into the sub-project level, yet data became aggregated at the PO level**. All sub-projects followed a standardized format, where the objectives were specified, and the rationale of activities were properly reviewed and cleared by both PKSF and the World Bank. Every sub-project developed a separate ToC to document how the key activities and outputs led to expected outcomes with separate results frameworks to track progress. GIS-based area maps were prepared to locate each sub-sector, and notably, all GIS-based maps were made available on the project website. However, at project completion, regular M&E data had been aggregated by the POs, thereby preventing a more detailed analysis of some aspects of project achievements, particularly as it relates to analyzing the portfolio of micro-loans to perform more sophisticated economic analyses.

61. **The PMU arranged for both on-site and off-site monitoring.** PKSF own comprehensive M&E system proved to be a useful and effective tool for keeping project implementation in check. Within this system, activities at the sub-project level were monitored by the PMU both on-site and off-site. From the PKSF side, on-site monitoring was done through the PMU of SEP, the Program and Audit department of PKSF and by an external audit firm. From the PO side, the on-site monitoring was done by the PIU of the PO, the Head Office of the PO, the Internal Audit Department and by an external audit firm. Off-site monitoring was realized by checking reporting systems from branch to the Head Office and from Head Office to the PKSF level. As part of regular field level monitoring, documentation and reporting progress, the POs adopted different environmental monitoring formats/forms/sheets such as borrower profiles, environmental baseline screening forms, promissory notes, environmental baseline images, environmental progress monitoring sheets, MIS reporting forms, etc. Finally, PKSF arranged for a number of studies to be conducted to assess project performance, including an Interim Assessment, a Rapid Assessment, a Mid-term Evaluation, a Final Evaluation, and 22 sectoral & sub-sectoral studies.

M&E Utilization

62. **M&E data and information was actively used for project management and accountability purposes.** As a project management tool, M&E was regularly discussed in sessions with the Technical Committee and the Steering Committee and used to take corrective action, such as an extension of the project closing date and the preparation of a Phase-two project to build on the early success of the SEP project. This allowed for the utilization of M&E data to track the project-implementation progress and performance.

Justification of Overall Rating of Quality of M&E

63. **The quality of M&E is rated Substantial**. The M&E system provided overall robust and reliable information on progress towards the achievement of the PDO. The RF was sufficient to assess the achievement of the objectives and track the results chain linkages, but it was not geared to assess project environmental outcomes (see also Section V). Furthermore, some M&E data collected by the POs could not be further disaggregated for analytical purposes at

the ICR stage. Yet, project M&E has contributed to the design of a scaled-up follow-on project and could provide important insights on advancing the agenda of green finance in the microenterprise sector.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

64. **The project was rated "Category B" (partial assessment) with expected significant positive environmental and social impacts.** Two safeguards policies were triggered: OP 4.01 Environmental Assessment as the project may involve small-scale construction to improve existing market and sanitation facilities and OP 4.10 Indigenous Peoples due to potential work in areas inhabited by indigenous peoples. The project design paid attention to pertinent social issues common in rural economies, which tend to be informal and family-based, including gender and child labor, through monitoring and prevention. Citizen engagement was emphasized, with feedback mechanisms, grievance redress mechanism, and third-party monitoring to ensure transparency and accountability.

65. **The project completed planned mitigation activities, including the timely preparation and disclosure of safeguards framework instruments, and safeguards implementation was rated satisfactory throughout implementation.** The project duly prepared and disclosed an Environmental Management Framework, a Social Management Framework, a Resettlement Policy Framework, and a Tribal People's Framework. While challenges arose, they were managed by the PIU to prevent negative impacts. The project itself facilitated MEs ability to manage solid waste, reduce emissions, manage effluents, etc. and therefore directly supported the compliance with safeguards policies and best practices. However, the PIU highlighted the need for further capacity building of Partner organizations, who lacked experience in implementing safeguards due diligence. The project did not conduct any activities in IP areas.

66. **Partner organizations arranged for extensive consultations, reaching more than 25,000 stakeholders.** Over 2,100 community/stakeholder consultations were conducted during sub-project proposal preparation and activity implementation by the POs. Participation lists show that more than 25,000 individuals participated in stakeholder consultations, of which women accounted for 38 percent. With their prior consent, women were also offered to be consulted separately, to ensure free expression of their voice and agency.

67. **Fifty-five grievance redressal committees (GRCs) were established at the local PO level, but no complaints were ever recorded**. The responsibility of ensuring that vulnerable individuals, MEs, and female entrepreneurs were aware of the project's Grievance Redress Mechanism and its functionality fell on the POs and their local chapters, where the sub-projects were implemented. However, throughout the implementation period, no complaints were lodged concerning land acquisition, resettlement, and displacement, or over other potentially negative impacts due to the detailed stakeholder consultations during project design.

Fiduciary Compliance

68. **The project remained in compliance with fiduciary requirements throughout implementation, and Fiduciary Management and Procurement were both rated Satisfactory at completion**. Throughout the project implementation PKSF maintained robust financial management team, which carried out all activities related to bookkeeping, accounting, reporting, disbursement in an efficient manner and in compliance with Bank requirements. PKSF prepared and submitted withdrawal applications seamlessly and was able to handle disbursement and documentation of expenditures in an efficient manner.

69. **Partner organizations functioned as financial intermediaries with oversight of MEs**. The three partner organizations functioned as financial intermediaries on-lending funds from PKSF to MEs. POs maintained separate bank accounts for sub-project funds and submitted monthly or bi-annual financial and physical progress reports and relevant bank statements to PKSF. None of the POs had access to an online financial portal, meaning all reporting and accounting of financial information was done manually, which is prone to delays and mistakes. To address the issue, a digital management information system was developed to capture real-time financial transactions. The SMART project is in the process of implementing the new system to establish an interface for POs to allow for expediting the verification of expenditures and replenishment from PKSF.

70. **Internal and external audits**: PKSF carried out annual internal audits of project and sub-project activities and expenditures incurred by PKSF itself and the POs to review compliance, accountability, and transparency of expenditures and the procurement process. Quarterly financial reports were regularly submitted to the Bank and required only minor revisions. All annual audit reports were timely submitted and with unqualified opinions.

71. **The project performed moderately well from a procurement point of view due to PKSF's solid experience with implementing similar Bank-funded projects.** Challenges mainly related to external factors, such as consultants' deliverables, POs' lack of experience and capacity, and the global pandemic. Quality assurance of deliverables sometimes required multiple rounds of reviews, which necessitated an extension of the contract validity duration. Inexperience at the level of POs with World Bank procurement standards required continuous hands-on support and training to execute day to day procurement activities. Due to the COVID-19 pandemic worldwide, initiation of many procurement packages were delayed and contract implementation in the field was hampered. There were no instances of mis procurement, and procurement audits were clean.

C. BANK PERFORMANCE

Rating: Satisfactory

Quality at Entry

72**.** **The project benefited from satisfactory quality at entry.** The project was based on a solid diagnostic of Bangladesh's development challenges and priorities, and it was closely aligned with the World Bank's country engagement strategy as well as its global agenda on environment and climate change. The project design was novel as the first environment project in the Bank to structure a microfinance on-lending program, and benefited from the expertise of a microfinance consultant, who formed part of the project preparation and implementation team. The design was ambitious without seeking to disrupt the microfinance sector by introducing too big of an agenda, but it overestimated MEs general readiness for adopting RECP processes and their need to first adopt OHS practices on the path towards environmental sustainability. Underpinned by solid M&E arrangements with independent verification of results, the project design incorporated lessons learned from the Bank's previous experience with PKSF, paid close attention to gender issues, and made adequate provisions for financial management, procurement, and safeguards, including timely preparation and disclosure of safeguards framework instruments. However, in hindsight the team could have paid more attention to the sub-project approval process, which was affected by implementation delay.

Quality of Supervision

73**.** **The Bank team provided adequate supervision guidance to the project through eleven supervision missions and a mid-term review with a frequency of two missions a year on average.** Despite COVID-19 disruptions, project disbursement consistently outpaced projections. With implementation well underway at the Mid-Term Review (MTR) and most targets overachieved, there was potential for revising targets upward in the Results Framework, however the team chose not to do so out of an abundance of caution due to uncertainty in the middle of COVID-19 pandemic. The Bank team maintained candid communication lines with both the PMU and Bank management, also during the pandemic when supervision was conducted virtually. This allowed issues to surface and be resolved and the team acted swiftly to extend the project closing date and ensure the proper completion of project activities. Furthermore, the team ensured smooth transition arrangements at project closure and capitalized on the opportunity to develop a new follow-on project aimed at sustaining the achieved outcomes.

Justification of Overall Rating of Bank Performance

74. **Overall Bank performance is rated Satisfactory. This is justified by the satisfactory quality at entry and satisfactory quality of supervision.**

D. RISK TO DEVELOPMENT OUTCOME

75. **The key risks to development outcomes relate primarily to maintaining the environmental practices introduced into the microenterprise sector through a favorable lending program:**

- *Sustainability of Practices:* Even after initial adoption, there is a risk that microenterprises may revert to non-environmentally friendly practices due to financial constraints or changing market dynamics, undermining the long-term impact on environmental sustainability.
- *Market Volatility:* Fluctuations in market demand for environmentally friendly products or services could affect the profitability and viability of microenterprises, potentially leading to reduced income generation and repayment capacity.
- *Policy and Regulatory Changes:* Shifts in government policies or regulations related to environmental standards or lending practices may impact the operating environment for microenterprises, affecting their ability to maintain compliance and access financing.
- *Technological Obsolescence:* Rapid advancements in environmental technologies could render the adopted practices obsolete over time, necessitating ongoing investment in upgrading or replacing equipment and processes to remain competitive and sustainable.
- *Climate-Related Risks:* Continued exposure to climate change-related hazards such as extreme weather events or environmental degradation may disrupt operations and infrastructure, posing risks to the resilience and viability of microenterprises in the long term.
- *Capacity Building and Institutional Support:* Without ongoing support and capacity building efforts, microenterprises may struggle to fully integrate and institutionalize environmental practices into their business models, limiting the scalability and replicability of successful initiatives.

76. **To mitigate these risks and safeguard development outcomes, the next-phase Sustainable Microenterprise and Resilient Transformation (SMART) project for US$250 million was approved on April 27, 2023**. Building on SEP, a Phase 2 project was approved to scale the impact of SEP and further anchor financing for environmental practices in the ME sector as a tool to reduce pollution. The SMART project leverages the institutional and contextual memory of existing SEP-PMU staff for project implementation. Going forward, it will be essential to prioritize ongoing monitoring and evaluation, provide targeted support for capacity building and market resilience, and foster collaboration with relevant stakeholders. Equally important will it be to develop the policy and institutional framework to govern financing for environmental practices in the ME sector by i.e., creating a registry

of microenterprises or a formal window for accessing this type of finance through government programs or private lenders.

V. LESSONS AND RECOMMENDATIONS

77. SEP has delivered several important lessons learned, some of which have already been incorporated into the next-phase SMART project.

78. **First, the experience of SEP has affirmed the importance of government-level support for the successful attainment of outcomes, particularly when the project scope is innovative in nature**. The GoB has shown unwavering support for the project given the close alignment of project objectives with government priorities. Innovative projects often face unique challenges that require extra attention and supportive policies and frameworks. As the first project of its kind in both the World Bank and PKSF, this was also true for SEP, which benefited from the dedicated team at PKSF and their direct support from the GoB. Future projects should be encouraged to tap into an agenda of top priority for the government and secure government support early in the planning stages. This can help ensure the best conditions for ongoing support through active involvement and endorsement from government bodies, lending credibility and legitimacy to the project, and mobilizing additional resources towards further investments and collaborations, as the experience of SEP has shown.

79. **Second, a number of smaller lessons of practical application have already informed the design and implementation of the SMART project, and may be relevant to other and similar projects**:

- ESP: Activities aimed at OHS and implementation of the ESF proved to be a gateway to introducing other environmentally sustainable practices in MEs through micro-financing mechanisms. Yet, in the new project, only activities related to RECP will be eligible for project financing to guide the adoption of more advanced ESPs.
- RF: In the SMART project, the adoption of more advanced ESPs is expected to have a measurable environmental impact. Therefore, the results framework in the SMART project includes indicators geared to assess environmental outcomes with four key environmental metrics.
- M&E: To allow for a more nuanced assessment of project outcomes, M&E under the SMART project has been designed to collect data at a more granular level. This includes geotagging financed activities and beneficiary MEs to understand the scope and scale of how beneficiary MEs may influence the adoption of ESPs among non-beneficiary MEs (the spill-over effect).

80. **Third, active and comprehensive dissemination of information on ESPs and on branded products through online platforms can be a good segue into digitalization of the value-chain to achieve further resource efficiency**. During project implementation, MEs and POs constantly posted and shared information on environmentally sustainable practices and products on social media. The continuous flow of information has opened new sales channels for MEs across various sectors and clusters. The emphasis on quality standards and green certification has propelled market reach and acceptance. In this way, digital integration has enhanced resource efficiency and enabled MEs to connect more effectively with customers and expand their market presence. Going forward it is recommended to build on this development and tap the potential in connecting MEs with customers and banking systems through e-commerce ecosystems and digital payment modalities in succeeding projects. This approach aligns with the broader goal of achieving sustainable and resource-efficient growth.

81. **Finally, collaborative efforts are necessary to manage the high costs of enforcing compliance with environmental standards**. Establishing a regulatory environment that incentivizes greener practices is vital for the growth of environmentally sustainable microenterprises. Policies targeted at MEs should be designed to encourage the adoption of cleaner technologies and practices, while going forward better enforcement policies will be needed to improve compliance.

ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS FRAMEWORK

PDO Indicators by Outcomes

To increase the adoption of environmentally sustainable practices by targeted microenterprises								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
Microenterprises targeted by the project that have adopted at least one environmentally sustainable practice (Number)	0.00	Dec/2017	20,000.00	Jun/2023	20,000.00	Feb/2024	48,845.00	Feb/2024
	Comments on achieving targets		Target 244 percent achieved. At appraisal, environmentally sustainable practices were defined as a business methodology or technology that would deliver resource efficiency, low pollution, and/or climate resilience. However, during implementation, eligible practices came to include occupational health and safety practices in order to meet the needs of MEs. Eligible practices were only defined during implementation, but it was originally intended that proof of adoption of ESPs would be assessed based on certification of applicable environmental quality standards. The target was defined as an adoption rate of 50 percent of the total number of microenterprises supported by the project. At completion, of the 65,124 MEs that have signed loan agreements, 75 percent have adopted at least one occupational health and safety measure or environmentally sustainable practice.					
Share of target beneficiaries with a rating of “satisfactory” or above on project interventions (Percentage)	0.00	Dec/2017	60.00	Jun/2023	60.00	Feb/2024	66.00	Feb/2024
	Comments on achieving targets		Target 110 percent achieved. Beneficiaries were defined to include the owners, family members, and workers in the microenterprises supported by SEP. In the Final Beneficiary Survey, a composite index was constructed pertaining to both financial and non-financial aspects of project interventions to measure beneficiaries satisfaction with SEP interventions. A combined 66 percent of beneficiaries report to be "Satisfied" or "Highly satisfied" on 11 different project dimensions, such as SEP financial services (e.g., soft loans and grants), service frequency, timeliness, utilization, as well as non-financial services like business development and market linkage, indicating a strong sense of approval and satisfaction with project interventions.					
	0.00	Dec/2017	30.00	Jun/2023	30.00	Jun/2023	65.00	Dec/2018

disaggregated by sex (Percentage)	Comments on achieving targets		Target 217 percent achieved.					
Targeted micro-enterprises that continue the adopted environmentally sustainable practice (Percentage)	0.00	Dec/2017	40.00	Jun/2023	40.00	Feb/2024	95.00	Feb/2024
	Comments on achieving targets		Target 238 percent achieved. The indicator tracked MEs interest in continuing to invest in adopted practices. The target was defined as 40 percent of MEs, who received a SEP-financed loan once and who continue the adopted environmentally sustainable practice for more than 12 months (the duration of the loan repayment period). The target was overachieved already by mid-term with over 70 percent of MEs expressing their intent to continue the adopted practice with own funds. The Final Beneficiary Survey reveal a sustained adoption of ESPs among MEs in both treatment (95 percent) and control (91 percent) groups, maintaining these practices for 12 months or more. There is a strong intent among MEs to continue with ESP adoption in the future, with a significant willingness to bear the associated costs independently, underscoring the perceived value of these practices. Health benefits emerge as the primary motivator for ESP adoption across both groups.					
disaggregate by sex of ME owner (Percentage)	0.00	Dec/2017	30.00	Jun/2023	30.00	Jun/2023	95	Dec/2018
	Comments on achieving targets		Target 316 percent achieved. The target was defined as 30 percent of female-headed MEs, who received a SEP loan and who continue the adopted environmentally sustainable practice for more than 12 months (the duration of the loan repayment period). At completion, the survey shows that 95 percent of women in the treatment group continue the ESP practices adopted with SEP funds, with a similar outcome for women in the control group who adopted environmental, health and occupational practices with own funds.					

Intermediate Indicators by Components

Enhancing Services and Enabling Systems								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
Micro enterprise clusters provided with common services (Number)	0.00	Dec/2017	25.00	Jun/2023	25.00	Feb/2024	25.00	Feb/2024
	Comments on achieving targets		Target achieved. In SEP, business clusters were recognized as a concentration of interconnected microenterprises situated within an adjoining geographical location. The indicator was defined as ME clusters with access to SEP-financed common services with a target of 25 revenue generating and 25 non-revenue generating common services established by the project in identified ME clusters. At project completion, the project has met the target by establishing common services of each kind in the identified ME clusters. The Final Beneficiary Survey shows that 33 percent of SEP-funded MEs report having accessed the Common Services compared to 25 percent in the control group, indicating a higher awareness of the presence and utility of common services among beneficiary MEs.					

Share of targeted microenterprises provided with capacity-building support to adopt environmental sustainable practices (Percentage)	0.00	Dec/2017	70.00	Jun/2023	70.00	Feb/2024	91.00	Feb/2024
	Comments on achieving targets	Target 130 percent achieved. The indicator was defined as the share of ME that received trainings or participated in exposure events. The target of 70 percent was identified against IRI 3. At completion, a total of 59,506 MEs have been trained. As a proportion of the MEs that have signed loans under the project (IRI 3), this is equal to 91 percent of all beneficiary MEs that have been reached with different capacity development support.						
of which female-owned enterprises (Percentage)	0.00	Dec/2017	30.00	Jun/2023	30.00	Feb/2024	52.00	Feb/2024
	Comments on achieving targets	Target 173 percent achieved. The indicator was defined as the share of women-owned MEs that have received training with a target of 52 percent. At closing, more than half the MEs that received training and capacity building support were women-owned.						

Strengthened Access to Finance for environmentally friendly and resilient microenterprises

Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
Microenterprises that sign loan agreements with PKSF under the project (Number)	0.00	Dec/2017	40,000.00	Jun/2023	40,000.00	Feb/2024	65,124.00	Feb/2024
	Comments on achieving targets	Target 162 percent achieved. The indicator was defined as an estimate of the number of MEs that would sign loan agreements to adopt environmentally sustainable practices with a target of 40,000 MEs. At completion, a total of 65,124 MEs had signed loan agreements under the Project. At the time of appraisal, it was estimated that loan amounts would vary between US$375 (for goat rearing) and US$7,500 (for light engineering/automobile workshop), with an average of US$2,325 per loan (including both IDA credit and PKSF contribution). With actual disbursements to ME loans under Component 2 of US$82 million (including both IDA credit and PKSF contribution), the actual average loan value at closing is US$1,336. This likely reflects the high demand for health and occupational safety measures and practices among MEs at the time of the global pandemic, which tend to be lower cost than i.e. the introduction of advanced production processes. It also compares well with the average loan size.						
of which female-owned enterprises (Percentage)	0.00	Dec/2017	30.00	Jun/2023	30.00	Jun/2023	84.00	Dec/2018
	Comments on achieving targets	Target 280 percent achieved. The indicator was defined as the proportion of women-owned MEs to have signed loan agreements under the project. At project completion, 84 percent of 65,124 beneficiary MEs were owned by women. The achievement compares well with PKSF's regular lending program with a portfolio of 1.3 MEs, in which 91 percent of loan takers are female (PKSF Annual Report 2022).						

B. KEY OUTPUTS

Outcome (i): to increase the adoption of environmentally sustainable practices by Targeted Microenterprises	
Outcome Indicators	OI 1: Microenterprises targeted by the project that have adopted at least one environmentally sustainable practice OI 2: Share of target beneficiaries with a rating of “satisfactory” or above on project interventions (disaggregated by gender) OI 3: Targeted micro-enterprises that continue the adopted environmentally sustainable practice (disaggregated by gender)
Intermediate Results Indicators	IRI 1: Micro-enterprise clusters provided with common services IRI 2: Share of targeted microenterprises provided with capacity building support to adopt environmentally sustainable practices (of which female) IRI 3: Microenterprises that sign loan agreements with POs under the project (of which female)
Key Outputs by Component (linked to the achievement of the Objective / Outcome 1)	***Component 1: Enhancing Services and Enabling Systems*** *Subcomponent 1.1: Support for Enabling Systems* a) Non-revenue-generating Physical Activities In each of the 25 ME clusters, SEP established different non–revenue-generating common facilities to enhance MEs growth, competitiveness and productivity. Facilities established included: - Waste management and sustainable farming - Enhancing dairy business - Sustainable water management - Brand development and market expansion - Community transformation and recognition See *Table 9* for a full list of non–revenue-generating common facilities established. See also Case Study 1 and 2 in Annex 8. b) Initiatives to increase eco-labeling and access to premium markets

	Three types activities were undertaken: increase capacity of MEs on eco-labeling and access to premium market, provide support for business and product certification, and brand development. Key outputs included: - A total of 1,413 training sessions were arranged to build MEs capacity on eco-labeling and access to the premium market, benefiting over 35,000 MEs (**Table** *6*). - A total of 1,246 MEs have received certificate/clearance from different authorities such as Department of Environment (DoE) and Bangladesh Standards and Testing Institution (BSTI). - A total of 11,973 MEs, or 25 percent of targeted participants, registered their businesses with relevant authorities to obtain trade licenses (Union Parishad/Pouroshava/City Corporation). - Created brand identity and developed related brand assets, such as brand name and logo development, packaging design and production support, for 37 POs in 15 different sectors. - MEs supported to formulate Standard Testing procedure and Standard Operating Procedure (SOP) as certification, quality assurance, and testing are key requirements for product placement in the market. - Communication products produced for brand development of MEs included documentaries, Online Video Commercials (OVCs), online news advertisements, social media campaign, billboards, shop signboards, posters. - Promoted all the products of the MEs under a single brand name 'SEP Suponno', which is recognized for its emphasis on safety, absence of harmful raw materials, and adherence to best practices. - See Case Study 3 and 4 in Annex 8. c) Capacity Building of POs - A total of 60 training sessions delivered to over 6,800 PO staff members to guide MEs in accessing finance and becoming more environmentally sustainable and climate resilient (**Table 12**). *Subcomponent 1.2: Capacity Development of the MEs* - Out of 65,124 MEs, a total of 59,506 MEs (52 percent female-owned) have received different types of awareness raising, capacity building and skills training (**Table** *5*). Training ranged from improved and safe production methods, livestock rearing, and aquaculture practices, as well as compost making, and personal safety norms. - See Case Study 5 and 6 in Annex 8. *Subcomponent 1.3: Investment in Revenue-Generating Common Services* - A total of 25 revenue-generating common facilities were established, one in each of the ME clusters (**Table** *10*). Services include, among other, food preparation, waste recycling, horticulture, dry fish processing, pottery making, and making of leather products.

Component 2: Strengthened Access to Finance for Environmentally Friendly and Resilient Microenterprises

- Of the 65,124 MEs supported, 75 percent (or 48,845 MEs) have adopted at least one of 20 defined business practices to protect the environment and make their business more resilient (according to the Final Beneficiary Survey).
- Combined, a total of 143,353 sustainable practices have been adopted by the 48,845 MEs, or approximately three practices on average per ME (**Table** *3*).

Component 3: Project Management, Communication and Knowledge Management, and Monitoring and Evaluation

a) Communication channels:

- The PMU maintained a website, a Facebook page, and a YouTube channel for SEP. The POs also maintained websites, Facebook pages, and YouTube channel. A total of 53 websites on sub-projects have been developed and 64 Facebook pages have been opened for each sub-project to disseminate the learnings, best practices, and achievements of SEP among the target audience.

b) Knowledge products:

- The PMU published (i) nine issues of the SEP Chronicles (Newsletter), (ii) four issues of Enfoque (theme-based publication), (iii) a booklet titled 'Towards Environmental Sustainability: A brief Account of 64 sub-projects of Sustainable Enterprise Project (SEP)', and (iv) a booklet titled 'Microenterprises Going Green: Selected Stories of Sustainable Enterprise Project (SEP)' on the 30 success stories from 30 sub-sectors.
- The POs published 64 brochures, 64 booklets, and numerous leaflets, and posters. They produced 64 baseline video documentaries, 64 end-line video documentaries, and countless digital content for social media.

c) Media engagement

- Recognized media outlets at the national level published more than 700 news articles on SEP and its sub-projects.
- Television channels telecast over 100 packages, including some full-length agricultural programs on SEP's sub-projects.
- SEP and its sub-projects have over 100,000 followers on Facebook, generated almost a million posts, and achieved over 100 million views.

d) Knowledge sharing

	- Eight collective intelligence workshops across the project are involving different stakeholders were organized to identify and facilitate learning opportunities to identify the problems behind adopting environmentally sustainable practices and find possible "nudging" solutions. - Four op-eds were published in national dailies to facilitate necessary advocacy based on the learnings of SEP. - One hundred cluster ambassadors were appointed and trained in self-reflexive dialogues to disseminate the awareness messages among fellow micro-entrepreneurs. - A national-level seminar titled "Environmentally Sustainable Practices in Microenterprises and Youth Engagement" was organized to address the challenges and solutions identified by microentrepreneurs. e) Microenterprise Fais 2024 - The "Shuponno Somahar: Environment-friendly Microenterprise Fair 2024" took place from 8-10 February 2024, at the Bangabandhu International Conference Center in Agargaon, Dhaka. - The fair featured exhibits from 47 Partner Organizations engaged in project implementation and 77 entrepreneurs who received brand development support, showcasing their products.

ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION

A. TASK TEAM MEMBERS

Name	Role	
Eun Joo Allison Yi	Team Leader	
Mirza Omer Baig	Financial Management Specialist	
Hasib Ehsan Chowdhury	Financial Management Specialist	
Nishat Noman	Procurement Specialist	
Bushra Nishat	Environmental Specialist	
Sabah Moyeen	Social Specialist	
Jorge Luis Alva-Luperdi	Counsel	
Shourov Kumar Sharma	Procurement Team	
Shahnun Nima Tania	Procurement Team	
Priyani Malik	Team Member	
Hosna Ferdous Sumi	Team Member	
Iqbal Ahmed	Team Member	
S. M. Zulkernine	Team Member	
Nina Tsydenova	Team Member	
Daniel Payares Montoya	Team Member	
Simon Bell	Team Member	
Poonam Rohatgi	Team Member	
Ruozi Song	Contributing ICR Author	ICR
Sanne Agnete Tikjøb	Main ICR Author	ICR

B. STAFF TIME & COST

Stage of Project Cycle	Staff Time & Cost	
	No. of Staff Weeks	US$ (including travel and consultant costs)
Preparation		
FY17	1.200	7,610.84
FY18	26.698	202,714.93
FY19	0.900	12,298.96
Total	28.80	222,624.73
Supervision/ICR		
FY19	22.433	140,053.54
FY20	19.551	130,875.49
FY21	25.996	164,888.34
FY22	36.601	212,860.41
FY23	20.126	140,672.53
FY24	14.075	114,880.72
FY25	1.000	1,894.80
Total	139.78	906,125.83

ANNEX 3. PROJECT COST BY COMPONENT

Components	Amount at Approval (US$M)	Actual at Project Closing (US$M)	Percentage of Approval (US$M)
Enhancing Services and Enabling Systems	24.00	21.64	90
Strengthened Access to Finance for environmentally friendly and resilient microenterprises	75.00	75.00	100
Project Management, Communication and Knowledge Management and Monitoring & Evaluation	11.00	6.56	60
Total	**110.00**	**103.20**	**94**

- The total budget of the project was US$130 millions of which IDA contribution was US$110 million and the rest amount of US$20 million was contributed by the PKSF.
- Due to the fluctuation of SDR and US$ rate, the total actual IDA contribution was app. US$104.82 million.
- Disbursement progress was approximately 98 percent as of March 31, 2024.

ANNEX 4. EFFICIENCY ANALYSIS

Project benefits

1. Through the provision of common services, loan disbursement, and the adoption of environmental sustainable practices, SEP generated both economic and environmental benefits. The project benefits estimated here are conservative results, as some of the benefits cannot be quantified. The breakdown of costs and benefits is summarized in Table 8.

2. The economic benefits capture MEs' profit generation under SEP. Through multifaceted interventions, SEP aimed to enhance the financial viability of MEs. The potential business growth could change both the sales revenue and expenses, particularly the cost of goods sold and operating expenses. As a result, the impact of SEP's interventions on MEs' financial performance is estimated by gauging the profits generated under SEP, the differences between the sales revenues and expenses, emphasizing its role in the economic empowerment of MEs in Bangladesh.

3. Estimating the changes in the sales revenue and expenses of all the 65,124 MEs induced by SEP requires collecting detailed financial information on each ME over the years, which is not feasible. Thus, this analysis relies on surveying a representative set of 5,887 MEs, covering all 11 sectors. Of these, 3,662 MEs were randomly allocated to the treatment group that received the intervention or benefit package provided by the SEP project, specifically the Agroshar loan. MEs in the control group did not receive any intervention but might still have had access to common services. The causal impact of SEP is evaluated through Difference-in-Differences (DiD), a direct comparison of outcomes between the treatment and control groups before and after the treatment measured in two rounds of the survey: one mid-term in 2021 and the other end-line in 2023. Additionally, Propensity Score Matching (PSM) is utilized to increase the comparability between the treatment and control groups.

4. According to estimates provided by PKSF, the sales revenue of the MEs in the representative sample has been significantly positively impacted by the SEP project. The yearly sales revenue increased by 249,635 BDT (2121.90 USD) between 2021 and 2023. Similar to sales revenue, expenses have also increased significantly by 97,004 BDT (824.53 USD) in the same period, implying an increase in yearly profit by US$1,297.37 for each ME on average. Assuming all the 65,124 MEs that received the Agroshar loan experienced the same profit increase, SEP contributes an increase in yearly sales revenue between 2021 and 2023 by US$138.19 million, an increase in yearly expenses by US$53.70 million, and thus an increase in yearly profits by US$84.49 million.

5. The environmental benefits only include the social benefits of avoiding carbon emissions through the management of cattle manure generated by dairy farms and beef fattening. The adoption of environmental sustainable practices can generate multiple environmental benefits, including resource efficiency, low pollution, and climate resilience, as clarified in Table 11 in Annex 7. However, the expected positive environmental impacts cannot be fully monitored and assessed due to the widespread and small-scale nature of project-supported activities, a limitation identified at the appraisal stage. Only the social benefits of avoiding carbon emissions in dairy farms and beef fattening can be quantified based on a back-of-the-envelope calculation using the amount of cattle manure managed, as provided by PKSF.

6. In 2023, 144,504 metric tons of manure were managed from dairy farms and beef fattening under SEP, as documented in the Waste Inventory by PKSF. One wet metric ton of manure creates 23 cubic meters of biogas, 65 percent of which is methane. One cubic meter of methane is equivalent to 0.0199958 metric tons of carbon emissions. Then, 144,504 metric tons of manure would generate 43,197.62 (144,504*23*65percent*0.0199958) metric tons of carbon emissions. The calculation of the social benefits of carbon reduction follows the guidance note on the shadow price of carbon in economic analysis, where the shadow price of carbon (SPC) ranges from US$38 per metric ton of

CO_2 equivalent to US$77 per metric ton in 2018. The social benefits of reducing these carbon emissions would be US$1.64 million and US$3.33 million under low and high shadow prices of carbon respectively. Assuming the amount of manure managed was the same every year throughout the project period, the yearly social benefits of carbon reduction under SEP range from US$1.64 million and US$3.33 million.

Project costs

7. Project costs incorporate the costs of all three components of the project, including contributions from both IDA and PKSF. Specifically, these costs include (1) enhancing services and enabling systems (US$21.64 million), (2) strengthened access to finance for environmentally friendly and resilient microenterprises (US$93.00 million), and (3) project management, communication and knowledge management, and monitoring and evaluation (US$8.56 million). All the loans disbursed were assumed to circulate by POs and treated as costs.

Cost-benefit analysis

8. The analysis indicates that the project had an IRR of 7 percent with an NPV of US$29.04 million without counting environmental benefits. When environmental benefits are included, the NPV ranges from US$37.06 million to US$45.30 million, depending on the shadow price of carbon. To calculate IRR and NPV, several assumptions were made to account for the cash flow of SEP during the implementation period from 2018 to 2023. First, it was assumed that the costs of providing all three project components occurred at the beginning of the project. Second, loans were disbursed throughout the project period. It would be overly optimistic to assume that the estimated yearly economic benefits from access to loans, based on two rounds of surveys in 2021 and 2023 respectively, were realized every year. Therefore, it was conservatively assumed that the economic benefits were realized only after the mid-term survey. In addition, since waste management began at the start of the project, it was assumed that the social benefits of carbon reduction from managing cattle manure were realized every year throughout the project. Lastly, the discount rate was assumed to be 2.25percent.

9. At appraisal, an economic analysis of the project was conducted to estimate the NPV and IRR of the project. The NPV was estimated to be US$218 million at a 10 percent discount rate and IRR was estimated to be 66 percent in the PAD. Though the projections seem to be very different compared to the ones in the ICR at project closing, the major driver of the difference is the assumption of when the economic benefits were realized. In the PAD, economic benefits started to be realized in the second year of the project in 2019. However, in ICR, it was conservatively assumed that the economic benefits were realized only in 2022 and 2023 due to a lack of data collection before the mid-term survey in 2021. If this assumption in ICR is relaxed and economic benefits are counted since the second year of the project, as in the PAD, the IRR would be 63 percent with an NPV of US$179 million at a 10 percent discount rate excluding environmental benefits. The NPV would range from US$187 million to US$194 million when only the social benefits of avoiding carbon emissions through the management of cattle manure are included.

10. Aside from the timing assumption for the realization of economic benefits, other projections in the PAD align with the results in the ICR at project closing. In the PAD, the average loan amount was expected to range from US$375 to US$7,500. At project closing, with US$93.00 million disbursed among 65,124 MEs, the average loan size was US$1,428, which falls within the expected range in the PAD. Additionally, the PAD anticipated minimal additional costs for environmental practices, ranging from 0.03 percent to 1 percent of total costs. At project closing, various investments for the establishment of MEs, broken down by fixed and working capital, showed no significant differences between the treatment and control groups for all investments except for the business premise.

11. The IRR for this project is comparatively lower than some other projects in South Asia. This is partly because the calculations do not fully capture the environmental benefits and are based on conservative assumptions. For instance, a previous World Bank project in Bangladesh, the Clean Air & Sustainable Environment project closed in 2019, achieved

an IRR of 33.14 percent. This project focused on reducing air pollution by promoting cleaner technologies among brick enterprises, with various social benefits such as health cost savings and reduced CO2 emissions factored into the calculation. Similarly, another World Bank project in India closed in 2019, the Chiller Energy Efficiency project, reported an IRR of 28.9 percent. This project aimed to enhance energy efficiency by replacing less efficient centrifugal chillers with more efficient models, and its IRR was estimated based on anticipated energy savings over a 20-year period rather than just during the project's duration.

Table 8: Estimated costs and benefits of the project during project completion

Cost-Benefit Analysis	
Item	Amount (million)
Costs	
Component 1: Enhancing Services and Enabling Systems	US$21.64
Component 2: Strengthened access to finance for environmentally friendly and resilient microenterprises	US$93.00
Component 3: Project Management, Knowledge Management, and Monitoring and Evaluation	US$8.56
Benefits	
Economic benefits	
ME's yearly sales revenue	US$138.19
ME's yearly expense (cost of goods sold and operating expenses)	- US$53.70
Environmental benefits	
Yearly social benefits of carbon (low value of SPC)	US$1.64
Yearly social benefits of carbon (high value of SPC)	US$3.33
Net benefits	
NPV (without SPC)	US$29.04
NPV (with the low value of SPC)	US$37.06
NPV (with the high value of SPC)	US$45.30
B/C ratio (without SPC)	1.24
B/C ratio (with the low value of SPC)	1.31
B/C ratio (with the high value of SPC)	1.38
IRR (%, without SPC)	7%
IRR (%, with the low value of SPC)	9%
IRR (%, with the high value of SPC)	11%

ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

The draft ICR was shared with the Borrower for comments on July 23, 2024. PKSF sent written comments to the ICR team on August 15, 2024. The comments are summarized below in combination with World Bank responses.

1. **Financial Data:** PKSF suggested adjusting financial figures throughout the report, including financing amounts, project costs, and disbursement figures for each component. However, this data is recovered from World Bank internal database and cannot be changed to ensure consistency between the report and figures reported by the World Bank.

2. **Project Outcome Rating:** PKSF suggested that the project's efficacy rating should be upgraded due to exceeding targets and achieving objectives. PKSF also advocated for a "High" rating for project efficiency, citing robust economic analysis. While preparing the report, the task team thoroughly followed the World Bank evaluation methodology and did not find enough evidence for a “High” rating of project outcomes.

3. **Monitoring and Evaluation:** PKSF suggested that project M&E should be rated “Highly Satisfactory”. PKSF responded that the M&E design was based on the results framework at appraisal and cited the lack of specific data requirements to capture environmental outcomes. While preparing the report, the task team thoroughly followed the World Bank evaluation methodology and did not find enough evidence for a “High” rating of overall M&E design, implementation, and utilization.

4. **Editorial:** PKSF suggested editorial in-text revisions to text and figures throughout the report. Those editorial comments have been incorporated into the final draft to the extent possible.

ANNEX 6. SUPPORTING DOCUMENTS

TABLE 1: KEY WORLD BANK PROJECT DOCUMENTS

Document	Report No.	Date
Project Appraisal Document	PAD2543	February 28, 2018
Financing Agreement - Strategic Climate Fund		May 16, 2018
Implementation Status and Supervision Reports		
Restructuring Paper	RES54760	March 20, 2023
Country Partnership Framework FY16-FY20	141189-BD	January 7, 2020
Country Partnership Framework FY23-FY27	181003-BD	April 5, 2023
SMART Phase 2 PAD	PAD5168	April 4, 2023
Environmental and Social Framework (ESF)		October 1, 2018

TABLE 2: KEY BORROWER PROJECT DOCUMENTS

Document	Date
Borrower's Project Completion Report	May 2024
Final Beneficiary Satisfaction Survey	March 2024
PKSF Annual Report 2022	July 2023
SEP Quarterly Progress Report (Q3)	September 2023

ANNEX 7. SUPPORTING EVIDENCE

TABLE 3: ELIGIBLE AND ADOPTED ENVIRONMENTALLY SUSTAINABLE PRACTICES (ESPs)

Practice	Description	# of actions adopted	RE	LP	CR	OHS
Practice 1	Health Safety and Use of PPE (gloves, safety glasses, mask, apron, boots)	28 638				x
Practice 2	First aid box	13 988			x	x
Practice 3	Fire safety management	3 739			x	x
Practice 4	Safe drinking water, hygienic toilet, hand washing facility	15 203				x
Practice 5	Air circulation system with proper ventilation at workplace. Set-up exhaust fan for microenterprises. Sufficient lighting at workplace.	7 389			x	x
Practice 6	Energy savings lighting, use of daylight by using transparent roof sheets, or installation of insulator with roof	5 717	x			
Practice 7	Remove overhead storage of the workers	827				x
Practice 8	Arrange resting place / feeding area for workers	2 672				x
Practice 9	Use solar panels for renewable energy use	2 437	x			
Practice 10	Reduce water pollution by different activities (making water containing pits, chambers, filtration system, improved drainage system etc.)	5 404		x		
Practice 11	Safe production process, safe inputs, safe packaging, safe transportation system, quarantine for ill or contaminated products/ animals, introduce safe storage for finished goods or inputs (fish or animal feeds)	10 320	x	x	x	
Practice 12	Use natural or organic inputs (fertilizer/dye) for safe production or packaging	6 603		x		
Practice 13	Activities to reduce air pollution/ odor	3 833		x		
Practice 14	Activities to reduce noise pollution	471				x
Practice 15	Activities to support to manage waste	15 751		x		
Practice 16	Activities only for climate change adoption	826			x	
Practice 17	Sign, symbol or posters on awareness (Reduce air pollution, noise pollution, water pollution, fire safety management, no smoking, first aid box, safe drinking water, reduction of water pollution, use of PPE etc.)	14 921	x	x	x	
Practice 18	Activities for capacity building of MEs (trainings) under pollution reduction, resource efficiency and climate change	3 204	x	x	x	
Practice 19	Miscellaneous (improved machines, raw materials, technology, reduced risks)	726	x			
Practice 20	Improve electrical wiring for micro-enterprises, specially manufacturing sub-sectors.	684	x			x
Total practices adopted		**143 353**				

RE: Resource efficiency - LP: Low pollution - CR: Climate resilience - OHS: Occupational health and safety.

TABLE 4: TRAINING AND CAPACITY BUILDING DELIVERED TO POS

Training Category	No. of Training Sessions	No. of PO Participants
Accounts, Finance, and Procurement	4	4 259
Agriculture, Livestock, and Aquaculture	8	337
Branding, Marketing, and Quality Standards	11	258
Communication, Reporting and	9	454
Different methodology of Survey	2	150
Environment and Social Safeguard Policy	23	1 279
Manufacturing	1	53
Others	2	47
Grand Total	**60**	**6 837**

TABLE 5: TRAINING AND CAPACITY BUILDING OF MES

#	Training	Male	Female	Total
1	Production Method (e.g., Fodder Production)	2,487	2,264	4,751
2	Safe Production (Mango, Banana, Pineapple,	5,625	8,224	13,849
3	Enterprise Development (Loom, Mini-garments)	3,720	4,318	8,038
4	Livestock Rearing (Buffalo, Cow, Poultry)	3,636	6,686	10,322
5	Technology Orientation (e.g., Fish Culture	3,915	3,315	7,230
6	Environmental Awareness in Business	277	455	732
7	Personal Safety Norms	3,056	1,566	4,622
8	Good Aquaculture Practice (GAqP)	2,665	2,320	4,985
9	GMP, HACCP, GAP	1,113	1,065	2,178
10	Compost Making (Vermi Compost)	291	392	683
11	Linkage Meeting	1,212	290	1,502
12	Others	510	104	614
	Total MEs trained	**28,507**	**30,999**	**59,506**

TABLE 6: ME CAPACITY BUILDING ON ECO-LABELING AND ACCESS TO PREMIUM MARKET

Training Topic	No. of Training	Male	Female	Total
Access to premium market	217	3 272	1 956	5 228
Business Certification	206	3 386	2 836	6 222
Product Certification	194	3 154	4 215	7 369
Environmental Certification	312	4 728	3 119	7 847
Environment Management	484	5 106	6 802	11 908
Total	**1 413**	**19 646**	**18 928**	**38 574**

TABLE 7: THE SATISFACTION LEVEL OF BENEFICIARIES WITH THE SEP IS DETERMINED THROUGH A COMPOSITE INDEX, WHICH IS BASED ON 11 PERCEPTION SURVEY INDICATORS

Level of Satisfaction	Beneficiary Group		
	Male (%)	Female (%)	Total (%)
Not satisfied	5	7	6
Somewhat satisfied	8	8	8
Moderately satisfied	21	20	21
Satisfied	36	33	34
Highly satisfied	31	32	31
Sample size (#)	**1 994**	**1 663**	**3 663**

TABLE 8: SECTOR-WISE PORTFOLIO OF SUB-PROJECT

Sectors	No of Sub-project
Agribusiness	**34**
Aquaculture	7
Horticulture	10
Livestock	14
Poultry	3
Manufacturing	**30**
Food Processing	9
Handicraft	3
Leather Processing and Shoemaking	2
Light Engineering	6
Metal Work	3
Mini Textile	6
Plastic Recycling	1
Grand Total	**64**

TABLE 9: LIST OF NON-REVENUE-GENERATING COMMON SERVICES UNDER SEP

Sl	Name of the Sub-sector	Under Non-Revenue Generating Physical Activities	Implementing PO	Location
1	Leather Product (Footwear)	• Cluster-based waste collection and management system • Community-based toilet and tube well	POPI	Bhairab, Kishoreganj
2	Dairy Farm	• Community-managed dairy firm linked drainage network system for waste management • Community-based bio-pesticide production unit • Common cow dung collection center	Unnayan Prochesta	Tala, Satkhira
3	Mini-Garments	• Design, product development and promotion service cum information hub	OSACA & PMK	Pabna Sadar & Narayanganj sadar
4	Beef Fattening	• Cattle lifting system (Ramp) in the cattle haat • Waste management with sanitation system in cattle market	DESHA	Mirpur, Kushtia
5	Salt Processing	• Multipurpose field level shed for workers with sanitation system	PMUK	Islampur, Cox's azar Sadar
6	Buffalo Rearing	• Climate-resilient buffalo shelter (Killa) with waste management and sanitation system	GJUS	Bhola Sadar
7	Vegetables	• Waste to compost production unit • Demonstration of ecological farming	Nobolok	Monirampur, Jashore
8	Pisciculture	• Construction of fish landing center with sanitation facilities	CCDA	Daudkandi. Cumilla
9	Horticulture (Banana)	• Waste management station: farm and market place	TMSS	Shibganj, Bogura
	Horticulture (Mango)	• Waste management point • Community latrine	PROYAS	Shibganj, Chapainawabganj
	Horticulture (Pineapple)	• Demonstration of ecological farming	SSS	Modhupur, Tangail
10	Loom	• Small-scale connecting road to market place	NDP & DABI	Belkuchi, Sirajganj & Adamdighi, Naogaon
11	Imitation Jewelry	• Design and product promotion service	SNF	Maheshpur, Jhenaidah
12	Jaggery Processing	• Improvement of sugarcane landing station with sanitation system	PCD	Khaschar, Pabna sadar

13	Poultry	• Poultry service center (PSC)	PIDIM Foundation	Mallikbari, Mymensingh
14	Food Processing (Rice Mill)	• Soil test for diagnosis of land ingredients	ESDO	Thakurgaon
15	Dairy Product	• Renovation of dairy product processing facilities with PPE for staff	FDA and	Charfesson, Bhola
16	Machinery & Equipment	• Model workshop development and demonstration • Foundry slag recycling demonstration.	GUK	Bogura sadar, Bogura
17	Pottery (Terracotta)	• Eco-friendly kiln development for pottery • Clay processing machine and relevant working shed for marginal community level • Develop model potter	Unnayan Prochesta	Kolaroa, Tala, Shatkhira & Paikgachha, Khulna
18	Eco-Friendly Construction Materials	• Innovative eco-product producing machines and Relevant Working Shed (Adobe block) • Develop model ME	ESDO	Thakurgaon Sadar, Ranishonkoil, Baliadangi, Thakurgaon Chirir Bandar, Dinajpur
19	Food Service (Street & Restaurant Food)	• Model ME development for demonstration	ESDO & PMUK	Bagura Sadar, Bagura & Rajshahi sadar, Puthia, Poba, Rajshahi, and Rangpur Sadar
20	Dry Fish Processing	• Rest and breast-feeding center • Organic waste management facilities	Coast Foundation	Cox's Bazar Sadar, Cox's Bazar
21	Plastic waste recycling	• Household plastic waste sorting and sorting system development • Technician for CNC dice cutting machine and recycling machine • Toilet • Plastic waste collection van	RIC	Dhaka Sadar, Dhaka
22	Automobile Workshop	• Hygienic Sanitation System; • Environment friendly model workshop	TMSS	Bagura sadar, Bogura & Naogaon Sadar, Naogaon
23	Handicraft	• Handicraft Display cum Promotion Center	SOPIRET	Noakhali sadar, Noakhali &

				Laxmipur Sadar, Raipur, Laxmipur
24	Floriculture	• WASH system and Compost chamber • Demonstration of drip irrigation system and ecological farming	RRF	Godkhali, Jhikargacha, Jashore
25	Electric Items	• Waste carrier van • Electric items information and service hub	BASTOB	Dania, Jatrabari, Dhaka

TABLE 10: LIST OF REVENUE-GENERATING COMMON SERVICES UNDER SEP

Sl	Name of the Sub-sector	Name of the Common Services	Implementing PO	Location
1	Leather Product (Footwear)	• Footwear common service center	POPI	Bhairab, Kishoreganj
2	Dairy Farm	• Quality fodder supply • Waste to vermicompost production center	Unnayan Prochesta	Tala, Satkhira
3	Mini-Garments	• Low sound generator	OSACA & PMK	Pabna Sadar & Narayanganj sadar
4	Beef Fattening	• Natural farming demonstration	DESHA	Mirpur, Kushtia
5	Salt Processing	• Alternative salt cultivation process by land development	PMUK	Islampur, Cox's azar Sadar
6	Buffalo Rearing	• High yielding fodder cultivation	GJUS	Bhola Sadar
7	Vegetables	• Quality input supplier	Nobolok	Monirampur, Jashore
8	Pisciculture	• Improved and quality input suppliers • Quality Fry/Fingerling Supply Unit	CCDA	Daudkandi. Cumilla
9	Horticulture (Banana)	• New Product development (Fiber from Banana Plants) • Waste to Compost Production Facilities	TMSS	Shibganj, Bogura
	Horticulture (Mango)	• Quality plant supply unit to introduce new and good varieties	PROYAS	Shibganj, Chapainawabganj
	Horticulture (Pineapple)	• Waste to trico-compost production unit • Safe pineapple producers' product promotion and sales center	SSS	Modhupur, Tangail
10	Loom	• Design and product development and promotion service center	NDP & DABI	Belkuchi, Sirajganj &

Sl	Name of the Sub-sector	Name of the Common Services	Implementing PO	Location
		• Low sound automatic power loom (demonstration)		Adamdighi, Naogaon
11	Imitation Jewelry	• Set up packaging machines	SNF	Maheshpur, Jhenaidah
12	Jaggery Processing	• Safe Jaggery production and processing plant	PCD	Khaschar, Pabna sadar
13	Poultry	• Biogas plant	PIDIM Foundation	Mallikbari, Mymensingh
14	Food Processing (Rice Mill)	• Full grain rice production	ESDO	Thakurgaon
15	Dairy Product	• Milking machine • Waste to vermicompost plant	FDA and	Charfesson, Bhola
16	Machinery & Equipment	• Modern Machineries (Vertical Machining Center)	GUK	Bogura sadar, Bogura
17	Pottery (Terracotta)	• Terracotta display center	Unnayan Prochesta	Kolaroa, Tala, Shatkhira & Paikgachha, Khulna
18	Eco-Friendly Construction Materials	• Modern machinery for common service [Introduce of programmable logic controller eco-block machine, eco-block, adobe blocks, hollow block, parking tiles]	ESDO	Thakurgaon Sadar, Ranishonkoil, Baliadangi, Thakurgaon Chirir Bandar, Dinajpur
19	Food Service (Street & Restaurant Food)	• Model street food cart • Establish environment-friendly food packaging mini-industries • Establish mini soap-making factories for using used oil	ESDO & PMUK	Bagura Sadar, Bagura & Rajshahi sadar, Puthia, Poba, Rajshahi, and Rangpur Sadar
20	Dry Fish Processing	• Portable cold storage for dry fish • Safe dry fish outlet	Coast Foundation	Cox's Bazar Sadar, Cox's Bazar
21	Plastic waste recycling	• Computer aided CNC machine for dice cutting • Technical support to plastic waste processing and production units (raw materials) from segregated plastic waste (non-food graded)	RIC	Dhaka Sadar, Dhaka

Sl	Name of the Sub-sector	Name of the Common Services	Implementing PO	Location
		• Improve existing plastic waste recycled factory to produce plastic tiles, roof tiles, bricks • Improve community-based plastic waste collection and segregation unit		
22	Automobile Workshop	• Modern machineries introduction [Car Lift, Car Spray Booth, Car Washer, Car dent fixing tool]; • Automobile workshop tools, Automotive wheel alignment System, ATF exchanger & Cleaner, Injector tester & Cleaner; • AC service station	TMSS	Bagura sadar, Bogura & Naogaon Sadar, Naogaon
23	Handicraft	• Nursary development using cocodust • Coconut shell handicrafts production (e.g. ornaments, spoon, mug, souvenier etc.) • Coconut coir / rope production in the local facility • Virgin coconut oil production	SOPIRET	Noakhali sadar, Noakhali & Laxmipur Sadar, Raipur, Laxmipur
24	Floriculture	• Model input supplier • Flower outlets	RRF	Godkhali, Jhikargacha, Jashore
25	Electric Items	• Modern Machineries and Services (Vertical Machining Center VMC)	BASTOB	Dania, Jatrabari, Dhaka

ANNEX 8. BENEFICIARIES

The project targeted commercially viable microenterprises primarily from the manufacturing and agribusiness sectors, encompassing various industries such as aquaculture, horticulture, livestock, poultry, food processing, leather processing & shoemaking, metal work, mini textile, handicrafts, light engineering, and plastic recycling. Loans were extended to informal microenterprises, acknowledging that formal registration practices in Bangladesh were limited, particularly among female entrepreneurs who were more likely to be informal operators.

Direct beneficiaries include ME owners who participated in training programs, received financial support, or accessed common services provided by SEP. Additionally, household members and employees associated with these MEs also benefited indirectly from the project's activities. Furthermore, the project's emphasis on promoting environmentally sustainable practices and enhancing business resilience had broader indirect benefits for the wider community and environment.

The annex has two sections: a) Key impacts on project beneficiaries as reported in the Final Beneficiary Survey, and b) case studies developed by PKSF reflecting some of the impacts that SEP has had on the ground.

a. Key impacts on project beneficiaries as reported in the Final Beneficiary Survey

Findings from the Final Beneficiary Survey underscore the significant positive impact of MEs ESP adoption on 11 dimensions of well-being, with particular emphasis on the empowerment of female entrepreneurs. Below are three of graphs that show beneficiaries responses to the impact of the project on livelihoods and employment, access to education, and access to healthcare. Responses to the other eight dimensions of well-being are similar to first three even though their graphs are not shown below. Those additional dimensions include impacts on household production and income, annual savings, family’s purchasing capacity, assets and investments, consumption and quality of life, psychological well-being, and social well-being (source: Final Beneficiary Survey).

TABLE 11: RESPONSE TO "HOW HAS YOUR INVOLVEMENT IN THE MICROENTERPRISE AFFECTED YOUR LIVELIHOOD AND EMPLOYMENT STATUS?"

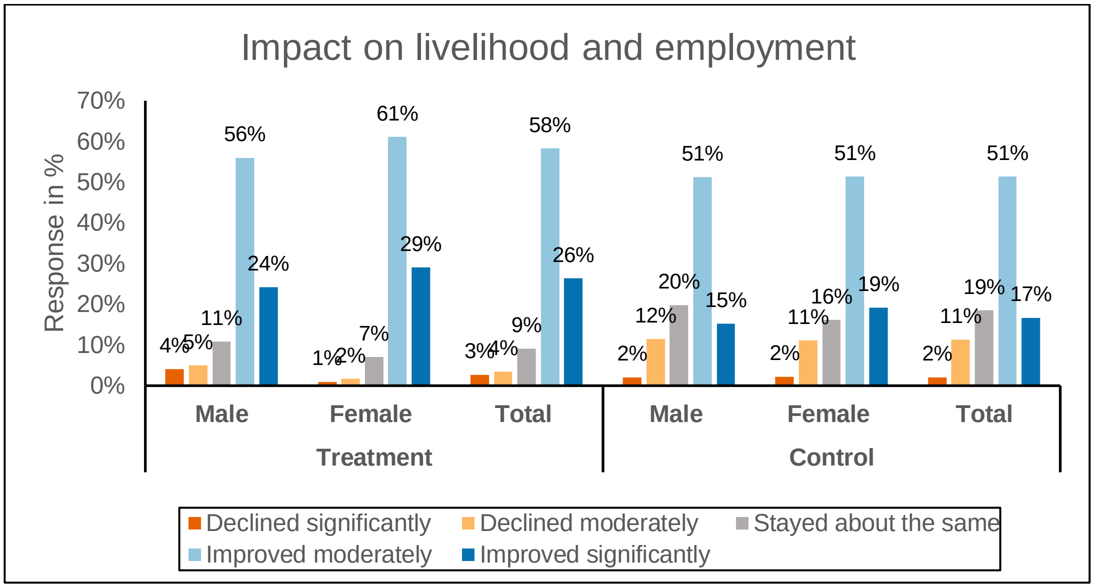


TABLE 12: RESPONSE TO " HOW HAS THE MICROENTERPRISE AFFECTED YOUR ACCESS TO EDUCATION OR EDUCATIONAL OPPORTUNITIES FOR YOUR FAMILY MEMBERS?"

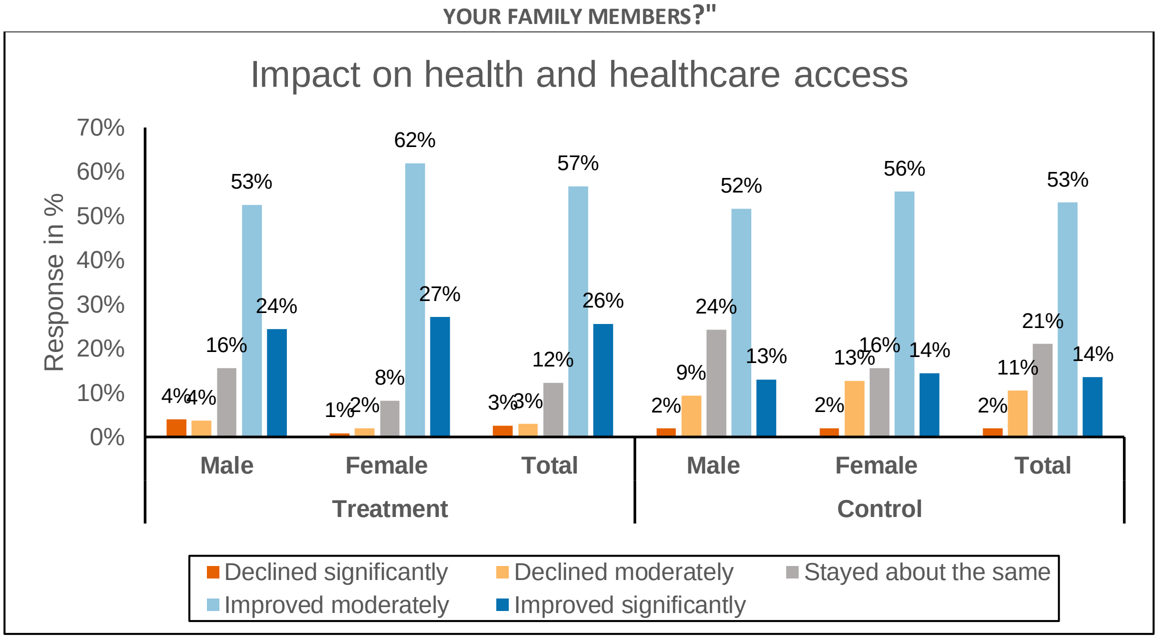
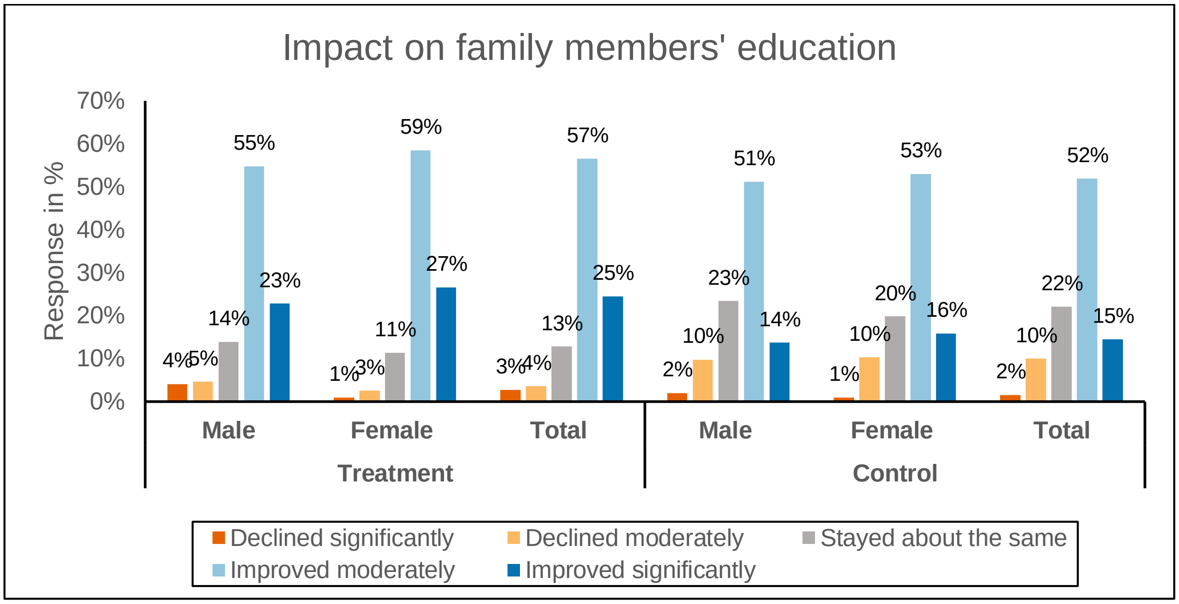


TABLE 13: RESPONSE TO "HOW HAS THE MICROENTERPRISE INFLUENCED YOUR OVERALL HEALTH AND HEALTHCARE ACCESS?"



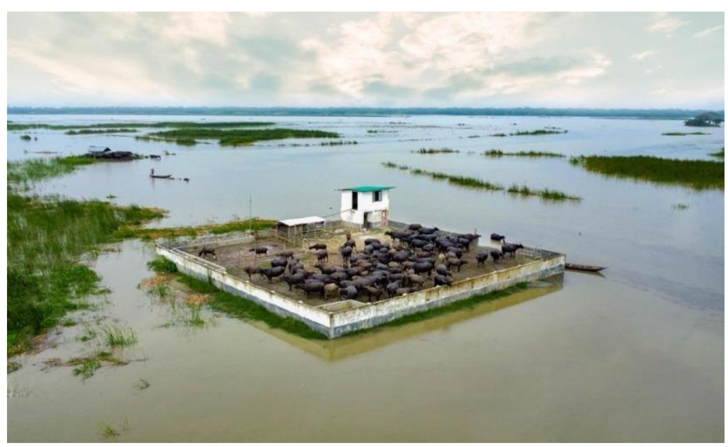
b. Case studies developed by PKSF reflecting project impacts

- Case Study 1: Killa protects buffalo farmers from cyclone
- Case Study 2: Transformative impact of SEP interventions on the Jiyala dairy cluster
- Case Study 3: From offline to online- Bhairab shoe cluster marches forward
- Case Study 4: Ataur Rahman's journey from village entrepreneur to national Dairy Icon
- Case Study 5: Climate-smart farming boosting livelihood and beating drought
- Case Study 6: Empowering Bhairab's shoe industry: A leap towards modernization
- Case Study 7: Turning waste into wealth: The sole revolution
- Case Study 8: From entrepreneur to environmentalist: Mini ETP in loom cluster provides sustainable solution to environment pollution

Case Study 1: Killa protects buffalo farmers from cyclone

The island district of Bhola is renowned for its buffalo milk curd, a staple dessert. Thousands of buffalos graze the chars during the day and return to temporary shelters, known as Killas, by sunset. These shelters are often submerged, even during low tides and rain, endangering the buffalos and causing losses from drawning and diseases. Furthermore, buffalo herders face harsh conditions without safe housing, potable water, cooking facilities, or lighting.

To address these challenges, Grameen Jaan Unnayan Sangstha (GJUS), a partner of PKSF, built a special Killa under the Sustainable Enterprise Project. This Killa, raised six feet above ground, includes separate areas for mother and baby buffalos, waste management facilities, lighting, and other amenities. It also provides safe shelters for herders, deep tube wells for potable water, hygienic toilets, kitchens, and solar power.



During a recent climate event (Cyclone Yaas), the new Killa saved hundreds of buffalos and has been positively received by buffalo farmers, serving as a model for modern Killas.

Case 2: Transformative impact of SEP interventions on the Jiyala dairy cluster: Reviving dignity and sustainability

The Jiyala village, along with Tala and Kalaroa areas, faced severe environmental and economic challenges due to mismanagement of cow dung and urine, which polluted soil, water, and air. Farmers struggled with market constraints and limited resources. The Sustainable Enterprise Project (SEP) transformed the community by introducing comprehensive waste management facilities and supporting farmers financially and technically.

Key Interventions and Impacts:
- Waste Management and Sustainable Farming: SEP helped manage waste from over 300,000 cows, producing 199,290 tonnes annually. This waste was utilized for vermicompost production, reducing pollution and enhancing soil fertility.
- Enhancing Dairy Business: SEP provided training and support to farmers, increasing milk production, profits, and improving animal health. Adoption of modern technology reduced feed costs and boosted productivity.
- Sustainable Water Management: SEP revived canals and water reservoirs by refining waste management systems, improving water quality and safeguarding groundwater. Environment Clubs and farmer associations were established for community-driven sustainability.
- Brand Development and Market Expansion: SEP enabled farmers to diversify products, enter premium markets, and establish vermicompost businesses, gaining consumer trust.
- Community Transformation and Recognition: The Jiyala dairy cluster, now a 'zero-waste' cluster, received Environment Clearance for 25 entrepreneurs. The community's reputation improved, becoming a model for environmentally friendly practices.

Overall, SEP significantly uplifted the Jiyala dairy cluster's economic and environmental status, empowering micro-entrepreneurs and serving as an inspiring model for other communities.

Case 3: From offline to online- Bhairab shoe cluster marches forward

The modernization of the footwear industry began in the late 1980s. At that time, district-based local cottage footwear enterprises operated to meet limited local demand but were unable to enter the premium market due to the sub-standard quality of their products. Under the Sustainable Enterprise Project (SEP), People's Oriented Program Implementation (POPI) launched a sub-project in the leather processing and shoemaking cluster in Kishoreganj. This included the 'eco-labelling and access to premium market' initiative, featuring awareness programs, training, and workshops on environmental, business, and product certification. Given the limited capacity of micro-enterprises (MEs) to address environmental issues, SEP took bold steps to improve the cluster's environmental conditions, organizing awareness programs and forming an environmental forum. As a result, the cluster is gradually adopting eco-friendly practices.

Previously, the cluster produced shoes with designs largely copied from Chinese and Indian models. Now, with the introduction of a software-based design facility, hand-made shoes targeting niche markets are being manufactured, attracting interest from foreign buyers and paving the way for future exports. A significant development was the launch of the e-commerce site Bhairab Shoe Mart, revolutionizing market access by enabling nationwide online sales.

Case 4: Ataur Rahman's journey from village entrepreneur to national Dairy Icon

In Haliagati village, Sirajganj district, Ataur Rahman's journey is a testament to resilience, entrepreneurship, and sustainability. Ataur, one of five siblings, faced financial constraints that prevented him from receiving formal education. Undeterred, Ataur turned to dairy farming with his family's five cows, supplying milk to sweet shops in 2018.

Struggling with market understanding and quality control, Ataur's dairy business saw limited growth until 2021 when he joined the SEP-Dairy Farm project, supported by PKSF and implemented by the National Development Program (NDP). Through SEP, Ataur received training in cheese-making from Bangladesh Agricultural University and financial support, including a BDT 200,000 loan.

Inspired by SEP's emphasis on environmental sustainability, Ataur introduced eco-friendly practices in his factory, such as a drain system, fire extinguisher, netting, first aid provisions, and improved worker facilities. This focus on a better work environment boosted productivity and profitability. His factory now processes around 2,500 liters of milk daily, employing six workers with good salaries.

With SEP's support, Ataur obtained BSTI certification, environmental clearance, and a food license, earning local recognition and the 'Dairy Icon' award on 1 June 2023. Ataur plans to produce advanced, eco-friendly dairy products, contributing to national health and productivity, showcasing the power of sustainable entrepreneurship.

Case 5: Climate-smart farming boosting livelihood and beating drought

A group of 25 women in Rajshahi's Shibrampur village has revitalized a fallow pond, turning it into an eco-friendly, climate-resilient fish farming site. Led by Mina Rani, these women dedicate hours to fish farming and cultivating the pond dyke, enhancing food security and boosting family incomes.

Groundwater levels are low most of the year, making traditional cultivation difficult. Daily-wage workers, like Mina's husband, often lost income due to these challenges. The Sustainable Enterprise Project (SEP) introduced climate-resilient technologies, such as layer-based carp fattening, to improve fish production sustainably.

In July 2021, the group adopted these techniques after attending SEP's training sessions on Good Aquaculture Practices (GAP), environmental certification, and post-harvest management. They now share profits equally among the 25 members. In August 2021, they formed an environment club called 'Surjer Hasi,' saving BDT 100 weekly to improve socio-economic conditions and address financial emergencies collectively.

Case Study 6: Empowering Bhairab's shoe industry: A leap towards modernization

In Bhairab, Kishoreganj district, traditional craftsmanship has hindered the local shoe manufacturing industry's ability to meet formal buyers' standards. Despite theoretical knowledge, practical skills remain lacking, and reliance on outdated tools and design replication limits growth.

To address these challenges, PKSF, under the Sustainable Enterprise Project (SEP), supported POPI in establishing the Shoe Craft Common Service Center in Bhairab. This 6,000-square-foot facility houses advanced machinery and offers modern skill development training, including services like beam cutting, laser cutting, and sewing. It also includes a modern shoe design and product development section with Shoe Master software, providing crucial hands-on design experience.

The center has become a focal point for quality shoe production, attracting local micro-entrepreneurs with its 24 modern machines, skilled operators, and affordable services. In the past 1.5 years, it has served 264 micro-entrepreneurs, emphasizing not only production skills but also environmental awareness, workplace safety, and regulatory adherence. In this way, SEP has transformed Bhairab's traditional shoe industry into a hub of modern, sustainable, and high-quality production.

Case Study 7: Turning waste into wealth: The sole revolution

Md. Jasim Uddin's journey in Bhairab is a remarkable transformation from a shoe sole retailer to a pioneering figure in recycling. With guidance from a partner organization under SEP, Jasim launched his sole recycling venture in 2018. Starting with one set of machines, the venture has grown to include five injection molding machines, three milling machines, and two grinding machines, employing 30 men and 7 women.

Jasim's eco-friendly approach focuses on utilizing PVC, PU leather, PU sole, and rubber waste from footwear production, supplemented by a small amount of virgin material. He processes 1.5 tonnes per day (40-45 tonnes monthly) of recyclable plastic waste sourced nationwide. The recycling unit, operating 24/7, produces 300 dozen soles daily, priced between BDT 80 to BDT 150 per dozen. These soles serve footwear entrepreneurs across districts like Bhairab, Rangpur, Mymensingh, and Chittagong.

The production process involves careful waste sorting, cutting, milling into powder, and injection molding into soles, emphasizing innovation and environmental responsibility. Regular worker training ensures efficient operations and a safe working environment. Despite market challenges, the recycling unit generates monthly sales of BDT 5-6 lakh, with profits ranging from BDT 40,000-50,000. Jasim plans to modernize further, collaborating with government and private sectors to expand waste utilization, create more jobs, and contribute to environmental conservation.

Case Study 8: From entrepreneur to environmentalist: Mini ETP in loom cluster provides sustainable solution to environment pollution

In Sirajganj, renowned for its vibrant handloom industry, SEP has initiated a remarkable transformation. Traditionally, Sirajganj's handloom sector has been a pillar of the local economy but has also inadvertently contributed to environmental pollution. Noise, soil, water, and air pollution have been prevalent issues, exacerbated by liquid effluents from dyeing and processing mills.

The initiative focused on enhancing the sustainability of MEs through the introduction of mini-ETPs (Liquid Effluent Treatment Plants), specifically designed to address environmental challenges. In collaboration with Aquatic Enterprise, the project developed mini-ETPs after extensive research and eighteen months of joint effort. These mini-ETPs now come in capacities of 3000, 5000, and 10000 liters per day, tailored to meet the needs and capabilities of micro-entrepreneurs.

Facing rising costs for drinking water due to liquid waste discharge, owners of dyeing and processing mills eagerly sought solutions. Entrepreneurs like Anwar Hossain Mollah from Tamai village became pioneers by investing BDT 7 lakh to set up the first mini-ETP. Encouraged by initial successes, other entrepreneurs such as Mehdi Hasan and Habibullah have also embraced the mini-ETP technology, driven by the SEP's support and funding from PKSF. However, the ongoing operation and maintenance of mini-ETPs, costing between BDT 6000 to BDT 16000 per month, pose financial challenges.

Government initiatives, support from the Department of Environment, and collaboration with private institutions have been crucial in promoting and sustaining mini-ETP adoption among dyeing and processing mill owners. Beyond its initial impact, the project has sparked broader interest, signaling a significant shift towards a more sustainable and environmentally conscious approach in the region's textile sector.